

Cryptocurrency Return Predictability: A Machine-Learning Analysis

Ilias Filippou¹, David E. Rapach^{*2}, and Christoffer Thimsen³

¹Washington University in St. Louis

²Federal Reserve Bank of Atlanta

³Aarhus University

February 25, 2024

Abstract

We investigate the out-of-sample predictability of daily cryptocurrency returns using modern machine-learning methods. We consider a large number of cryptocurrencies (41) and a rich set of predictors relating to network value and activity, momentum, technical signals, and online activity. We find that return predictability is an important feature of the cryptocurrency market: machine-learning methods significantly improve the statistical accuracy of cryptocurrency return forecasts and provide substantial economic value to investors. Predictors relating to momentum, size, and value stand out as important determinants of future cryptocurrency returns. Nonlinearities also play a significant role in improving cryptocurrency return predictability.

JEL classifications: C52, C53, G11, G12, G17

Keywords: Out-of-sample return prediction, Random forest, XGBoost, Deep neural network, Network value, Network activity, Time-series momentum, Investor attention, Shapley values

^{*}Corresponding author. Send correspondence to David Rapach, Research Department, Federal Reserve Bank of Atlanta, 1000 Peachtree Street NE, Atlanta, GA 30309; email: dave.rapach@gmail.com. The views expressed here are those of the authors and not necessarily those of the Federal Reserve Bank of Atlanta or the Federal Reserve System. Any remaining errors are the authors' responsibility. We thank conference and seminar participants at the 2023 Northern Finance Association Annual Conference, International Association for Applied Econometrics 2023 Conference, Autumn 2023 INQUIRE UK Residential Seminar, 2023 Australasian Finance and Banking Conference, Federal Reserve Bank of Atlanta, and Washington University in St. Louis, as well as Toni Braun, Andrew Detzel, Philippe Goulet Coulombe, Paul Freed (AFBC discussant), Nikolay Gospodinov, Yukun Liu, Jack Strauss, Larry Wall, Chengyu Zhang (NFA discussant), and Guofu Zhou, for valuable comments.

1. Introduction

Cryptocurrencies are an asset class with a relatively short and checkered history. They have experienced large swings in value, casting doubt on their ability to serve as new mediums of exchange, and their comparatively weak regulatory oversight raises concerns about market manipulation and fraud (e.g., [Griffin and Shams, 2020](#); [Flitter and Yaffe-Bellany, 2023](#)). Nevertheless, cryptocurrencies and their underlying blockchain technologies have the potential to profoundly affect the financial system and central banking ([Prasad, 2021](#)), and cryptocurrencies are now an important asset class for investors ([Harvey et al., 2022](#)). Concomitant with the advent and growth of cryptocurrencies, a burgeoning literature investigates asset pricing in the cryptocurrency market along various dimensions, such as arbitrage across exchanges (e.g., [Makarov and Schoar, 2020](#)), market equilibrium (e.g., [Cong et al., 2021](#); [Biais et al., 2023](#)), and empirical risk factors (e.g., [Liu et al., 2022](#)).

In this paper, we undertake a comprehensive analysis of the cryptocurrency market in terms of out-of-sample return predictability. Return predictability is a leading topic in asset pricing, as it relates to, among other things, information processing in financial markets, market efficiency, the determinants of time-varying expected returns, and dynamic asset allocation. We focus on out-of-sample tests, as such tests provide the most rigorous evidence regarding asset return predictability (e.g., [Goyal and Welch, 2008](#); [Nagel, 2021](#); [Martin and Nagel, 2022](#)). A handful of studies analyze out-of-sample cryptocurrency return prediction, focusing on Bitcoin; for example, [Huang et al. \(2019\)](#), [Detzel et al. \(2021\)](#), and [Gradojevic et al. \(2023\)](#) use technical indicators to predict daily and weekly Bitcoin returns, while [Cheah et al. \(2020\)](#) and [Chen et al. \(2021\)](#) predict daily Bitcoin returns using a variety of predictors. We extend the existing cryptocurrency literature by investigating the out-of-sample predictability of a large number of daily cryptocurrency returns using a broad array of predictors and a diverse set of modern machine-learning methods. We analyze both the statistical and economic significance of cryptocurrency return predictability. We

also explore the sources of return predictability, thereby shedding light on the economic mechanisms underlying cryptocurrency return predictability.

We begin with daily data for 48 cryptocurrencies from [CoinMetrics](#) for which prices and data for an extensive set of characteristics relating to network value and activity are available. After dropping “stable coins” and “wrapped tokens” that behave very differently from most cryptocurrencies, we have 41 cryptocurrencies. In addition to predictors based on network value and activity (i.e., cryptocurrency “fundamentals”), we construct market-based predictors relating to time-series momentum ([Moskowitz et al., 2012](#)) and price-based technical signals, which are popular with many practitioners. To consider measures of investor attention and sentiment, we also download cryptocurrency search-volume data from [Google Trends](#) as well as [Reddit](#) comments and [Factiva](#) articles relating to cryptocurrencies. Overall, we employ a rich set of predictors from manifold categories, which allows us to investigate a broad assortment of potential drivers of cryptocurrency return predictability.

We compute daily cryptocurrency return forecasts using a variety of machine-learning methods.¹ The first forecast is based on elastic net (ENet, [Zou and Hastie, 2005](#)) estimation of a linear model that includes all of the predictors (linear-ENet forecast). Conventional ordinary least squares (OLS) estimation of a large-scale linear model with correlated predictors is highly susceptible to in-sample overfitting. The ENet is an extension of the seminal least absolute shrinkage and selection operator (LASSO, [Tibshirani, 1996](#)) that helps to guard against overfitting by employing penalized regression to shrink the coefficient estimates toward zero. The remaining forecasts allow for nonlinearities in the prediction model, thereby creating scope for the forecasts to capture potentially important nonlinearities in the underlying data-generating process. We construct two forecasts based on decision trees: random forest ([Breiman, 2001](#)) and [XGBoost](#) ([Chen and Guestrin, 2016](#)). Decision trees partition

¹Machine learning is growing in popularity in finance; for example, a spate of recent studies use machine-learning methods to forecast stock returns, including [Chinco et al. \(2019\)](#), [Freyberger et al. \(2020\)](#), [Gu et al. \(2020\)](#), [Kozak et al. \(2020\)](#), [Dong et al. \(2022\)](#), and [Avramov et al. \(2023\)](#).

the predictor space into regions that allow for complex multi-way interactions among the predictors in producing the forecast. We also use a deep neural network to forecast cryptocurrency returns. Neural networks can approximate any smooth function (e.g., [Cybenko, 1989](#); [Funahashi, 1989](#); [Hornik et al., 1989](#); [Hornik, 1991](#)), so they accommodate general nonlinearities in predictive relations.

Because asset returns inherently contain a large unpredictable component, we are dealing with noisy data, which makes it vital to guard against overfitting when training the machine-learning models, especially given the large number of predictors that we consider. To help prevent overfitting, we tune multiple hyperparameters for each machine-learning model. Furthermore, to keep the fitted forecasting models timely, we re-tune the hyperparameters and re-train the models on a regular basis over time. We only use information available at the time of forecast formation when tuning the hyperparameters and training the models so that the forecasts do not entail “look-ahead” bias. In addition to forecasts based on the individual machine-learning methods, we consider ensembles that combine individual machine-learning forecasts. Ensembles are popular in machine learning. They recognize that we cannot know a priori the best forecasting model and allow for potential complementarities between the individual forecasts.

We find that daily out-of-sample cryptocurrency return predictability is empirically important, both statistically and economically. Indeed, the degree of cryptocurrency return predictability is quite striking. With respect to statistical accuracy, the machine-learning forecasts outperform the prevailing mean benchmark forecast in terms of out-of-sample mean squared error (MSE) in the vast majority of cases for the individual cryptocurrencies. The prevailing mean forecast assumes that returns are not predictable; given the intrinsically large unpredictable component in asset returns, it constitutes a stringent benchmark for analyzing out-of-sample return predictability (e.g., [Goyal and Welch, 2008](#)).² The machine-learning forecasts also significantly outperform the prevailing mean benchmark in terms of

²Our results are robust to an autoregressive benchmark forecast.

MSE for all of the cryptocurrencies taken together. Furthermore, the machine-learning forecasts that allow for nonlinearities (i.e., random forest, XGBoost, and deep neural network) are significantly more accurate than the linear-ENet forecast, so allowing for nonlinearities in fitted prediction models improves out-of-sample cryptocurrency return prediction. Ensemble forecasts also perform well. Large swings in value are a well-known feature of cryptocurrencies, and our machine-learning models are adept at improving return prediction around such swings.

In addition to improving statistical accuracy, the machine-learning forecasts are economically valuable. We consider a mean-variance investor who allocates between an individual cryptocurrency and risk-free Treasury bills. The investor realizes substantial utility gains by relying on machine-learning forecasts to guide asset allocation compared to ignoring return predictability by relying on the prevailing mean benchmark forecast. We also construct long-short portfolios that invest in multiple cryptocurrencies guided by the machine-learning forecasts. Specifically, we sort the available cryptocurrencies based on their return forecasts; the portfolio goes long (short) the cryptocurrencies with the highest (lowest) return forecasts. The long-short portfolios exhibit impressive performance in terms of Sharpe, Calmar, and Sortino ratios, and they typically perform substantially better than passively holding a value-weighted market portfolio of cryptocurrencies (as well as the value-weighted equity market portfolio). The long-short portfolios based on the machine-learning forecasts also generate statistically significant and economically sizable alphas in the context of the [Liu et al. \(2022\)](#) cryptocurrency three-factor model. With regard to economic value, we again find that nonlinearities are important, as long-short portfolios based on the nonlinear forecasts usually generate larger economic gains than those based on the linear-ENet forecast.

To glean insight into the sources of cryptocurrency return predictability, we use [Shapley \(1953\)](#) values to interpret the fitted machine-learning models that generate the return forecasts. According to Shapley-based variable-importance measures, among the most important predictors are time-series momentum, network value, the address-to-network value ratio, and

variables relating to network activity. These have notable parallels from other asset classes, including momentum, size, and value effects in the equity market. Investor attention via online activity, as captured by the number of Google Trends searches as well as Reddit comments and Factiva articles, also appears important. Shapley-value plots reveal significant nonlinearities in predictive relations for the fitted models, further highlighting the relevance of nonlinearities for predicting cryptocurrency returns. The momentum, size, and value effects are characterized by substantive nonlinearities in the context of Shapley-value plots. Shapley-value plots also indicate important nonlinearities for predictors based on network activity and investor attention. Overall, the interpretation of the fitted machine-learning models indicates that cryptocurrency return predictability is due to forces relating to a combination of rational time-varying risk premia and mispricing, where the latter can result from investor learning and limits to arbitrage (Shleifer and Vishny, 1997) in the cryptocurrency market.

The remainder of the paper is organized as follows. Section 2 presents the data. Section 3 describes the machine-learning methods used to generate the cryptocurrency return forecasts. Section 4 reports out-of-sample results in terms of statistical accuracy. Section 5 analyzes the economic value of the cryptocurrency return forecasts. Section 6 uses Shapley values to investigate the economic sources of cryptocurrency return predictability. Section 7 concludes.

2. Data

We use daily cryptocurrency prices in US dollars from CoinMetrics, a leading provider of cryptocurrency data. We begin with 48 cryptocurrencies for which prices and data for a large number of characteristics are available. Five of the cryptocurrencies are “stable coins” that are at least putatively pegged to the US dollar. Two more cryptocurrencies are “wrapped tokens” that are designed to trade 1:1 with Bitcoin. Because they behave quite differently from other cryptocurrencies, we drop the five stable coins and two wrapped tokens, providing

41 cryptocurrencies for our analysis.

2.1. Cryptocurrency Excess Returns

CoinMetrics furnishes US dollar closing prices (“reference rates”) at midnight coordinated universal time for every day of the year. Each price is a volume-weighted average of prices across exchanges that qualify as “constituent markets” according to criteria established by CoinMetrics.³ We compute the day- t return as the proportional change in the closing price from day $t - 1$ to t and compute the excess return using the daily risk-free return from CRSP.⁴ The cryptocurrencies come into existence at different dates, so data for different cryptocurrencies start at different dates. Our data extend through the end of 2021.

Table 1 reports summary statistics for the 41 cryptocurrency excess returns. Bitcoin is the oldest cryptocurrency, with a sample starting date of January 1, 2011, resulting in 4,018 daily observations. Litecoin is the next oldest, with a sample starting date of April 2, 2013 (3,196 observations). The youngest cryptocurrency in the sample is Internet Computer, which has a sample starting date of May 12, 2021 (234 observations). The fourth column reports market capitalization at the end of 2021. Two cryptocurrencies stand out in terms of market capitalization: Bitcoin (\$877B) and Ethereum (\$434B). The third largest market capitalization is for XRP (\$83B), which is around ten and five times smaller than those for Bitcoin and Ethereum, respectively. The vast majority of the cryptocurrencies in the sample have a market capitalization above \$1B at the end of 2021.⁵

The magnitudes of the annualized means and volatilities in the fifth and sixth columns, respectively, of Table 1 are much larger than those typically seen for assets such as equities, bonds, and currencies. The youngest cryptocurrency in the sample (Internet Computer)

³The list of constituent markets varies across cryptocurrencies.

⁴CRSP does not provide risk-free return observations for weekends and holidays. We fill in each missing value with the previous day’s value. Our out-of-sample forecasting results are similar if we use the simple instead of the excess return.

⁵The six exceptions are Balancer (\$0.82B), Decred (\$0.96B), HedgeTrade (\$0.71B), Livepeer (\$0.98B), OMGNetwork (\$0.83B), and UMA (\$0.97B).

Table 1. Summary statistics

The table reports summary statistics for daily excess returns for the cryptocurrency in the first column. Cryptocurrency returns are computed using US dollar daily closing prices from [CoinMetrics](#). The market capitalization in the fourth column is for the end of the sample (2021-12-31). The statistics in the fifth through tenth columns are computed after omitting the first 28 days of data.

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cryptocurrency	Sample start	Obs.	Market cap. (\$B)	Ann. mean (%)	Ann. vol. (%)	Min. (%)	Max. (%)	Ann. Sharpe ratio	Auto-corr.
linch	2020-12-27	370	3.59	140.18	154.59	-35.75	42.66	0.91	-0.01
Aave	2020-10-11	447	4.10	251.38	141.91	-31.05	30.81	1.77	0.01
Algorand	2019-06-23	923	16.69	125.62	135.42	-46.72	48.00	0.93	-0.06
Balancer	2020-06-26	554	0.82	137.34	149.89	-29.21	58.25	0.92	-0.03
Basic Attention Token	2017-10-07	1,547	1.82	137.00	133.09	-42.14	46.61	1.03	-0.09
Bitcoin	2011-01-01	4,018	876.86	152.45	96.30	-48.57	48.57	1.58	0.02
Bitcoin Cash	2017-08-02	1,613	8.17	77.74	131.19	-42.93	51.05	0.59	0.02
Bitcoin SV	2018-11-16	1,142	2.30	113.11	153.88	-43.81	144.08	0.74	-0.08
Cardano	2017-12-02	1,491	43.17	91.85	118.99	-38.89	34.00	0.77	-0.04
Chainlink	2017-09-30	1,554	19.62	207.95	143.81	-47.06	60.79	1.45	-0.05
Compound	2020-06-19	561	2.00	103.24	131.11	-25.72	23.05	0.79	-0.02
Crypto.com Coin	2019-03-21	1,017	55.95	133.21	113.52	-37.84	58.59	1.17	-0.09
Curve DAO Token	2020-08-16	503	8.97	245.86	188.96	-41.04	54.24	1.30	-0.03
Dash	2014-02-09	2,883	1.41	150.58	136.90	-37.78	107.00	1.10	-0.01
Decentraland	2017-08-26	1,589	7.19	334.39	245.93	-67.52	299.28	1.36	0.03
Decred	2016-05-18	2,054	0.96	155.39	135.59	-38.90	56.94	1.15	-0.11
Dogecoin	2014-01-24	2,899	22.62	177.98	181.45	-40.17	308.39	0.98	0.06
Ethereum	2015-08-09	2,337	433.79	193.48	117.34	-43.20	35.07	1.65	-0.04

has an annualized average excess return of -102.69% . The average excess returns are positive over their samples for the remaining cryptocurrencies, ranging from 72.32% (Neo) to 334.39% (Decentraland). The average excess return is above 100% (200%) for 31 (11) of the cryptocurrencies. All but two of the annualized volatilities in the sixth column are

Table 1 (continued)

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cryptocurrency	Sample start	Obs.	Market cap. (\$B)	Ann. mean (%)	Ann. vol. (%)	Min. (%)	Max. (%)	Ann. Sharpe ratio	Auto-corr.
Ethereum	2016-07-26	1,985	4.52	136.98	129.93	-39.80	74.66	1.05	-0.01
Classic									
FTX Token	2019-08-21	864	12.85	190.45	92.76	-24.99	32.21	2.05	-0.03
Gnosis	2017-05-03	1,704	5.31	94.52	126.45	-35.77	64.44	0.75	-0.03
HedgeTrade	2019-11-03	790	0.71	291.80	291.28	-40.24	258.78	1.00	-0.05
Huobi Token	2019-03-07	1,031	4.70	99.16	104.71	-38.70	60.96	0.95	-0.06
Internet Computer	2021-05-12	234	11.67	-102.69	151.67	-22.37	43.53	-0.68	0.05
Litecoin	2013-04-02	3,196	10.15	118.39	131.31	-42.64	110.30	0.90	0.01
Livepeer	2018-12-21	1,107	0.98	256.31	198.43	-53.37	139.58	1.29	-0.01
Maker	2017-12-27	1,466	2.31	88.92	124.99	-56.84	51.77	0.71	-0.13
Neo	2017-07-16	1,630	2.57	72.32	127.85	-38.77	45.75	0.57	-0.05
OMGNetwork	2017-07-16	1,630	0.83	94.08	139.88	-42.61	69.18	0.67	-0.05
Perpetual Protocol	2021-02-05	330	1.31	195.64	170.70	-25.37	39.46	1.15	-0.06
Polkadot	2020-08-21	498	30.21	218.37	137.93	-35.06	39.07	1.58	-0.03
Quant	2019-03-17	1,021	4.37	257.47	147.47	-42.70	53.63	1.75	-0.07
Stellar	2015-10-01	2,284	28.19	186.93	157.16	-33.96	101.07	1.19	0.10
SushiSwap	2020-09-02	486	2.26	305.43	171.79	-34.96	37.19	1.78	-0.09
Synthetix	2020-04-10	631	1.32	211.81	142.76	-34.18	25.62	1.48	-0.05
Tezos	2018-07-01	1,280	3.80	97.37	123.47	-44.25	32.88	0.79	-0.06
UMA	2020-09-09	479	0.97	144.20	161.04	-33.29	70.66	0.90	0.03
Uniswap	2020-09-19	469	17.09	244.94	146.43	-29.87	45.94	1.67	-0.08
XRP	2014-08-16	2,695	83.09	166.83	149.10	-47.09	174.22	1.12	-0.02
yearn.finance	2020-07-26	524	1.22	181.94	157.92	-34.51	46.65	1.15	0.05
Zcash	2016-10-30	1,889	1.76	87.99	123.96	-38.42	64.73	0.71	-0.01

above 100% and reach as high as 291.28% (HedgeTrade). Despite the high volatilities, the high average excess returns typically translate into sizable annualized Sharpe ratios in the ninth column. The Sharpe ratios are at least 0.57 for all of the cryptocurrencies, with the

exception of Internet Computer (-0.68), and over half (23) have Sharpe ratios of one or more. The Sharpe ratios for the two largest cryptocurrencies, Bitcoin and Ethereum, are 1.58 and 1.65, respectively. The minimum and maximum values in the seventh and eighth columns, respectively, clearly show that daily cryptocurrency returns are characterized by quite extreme values, while the autocorrelations in the last column are small in magnitude.

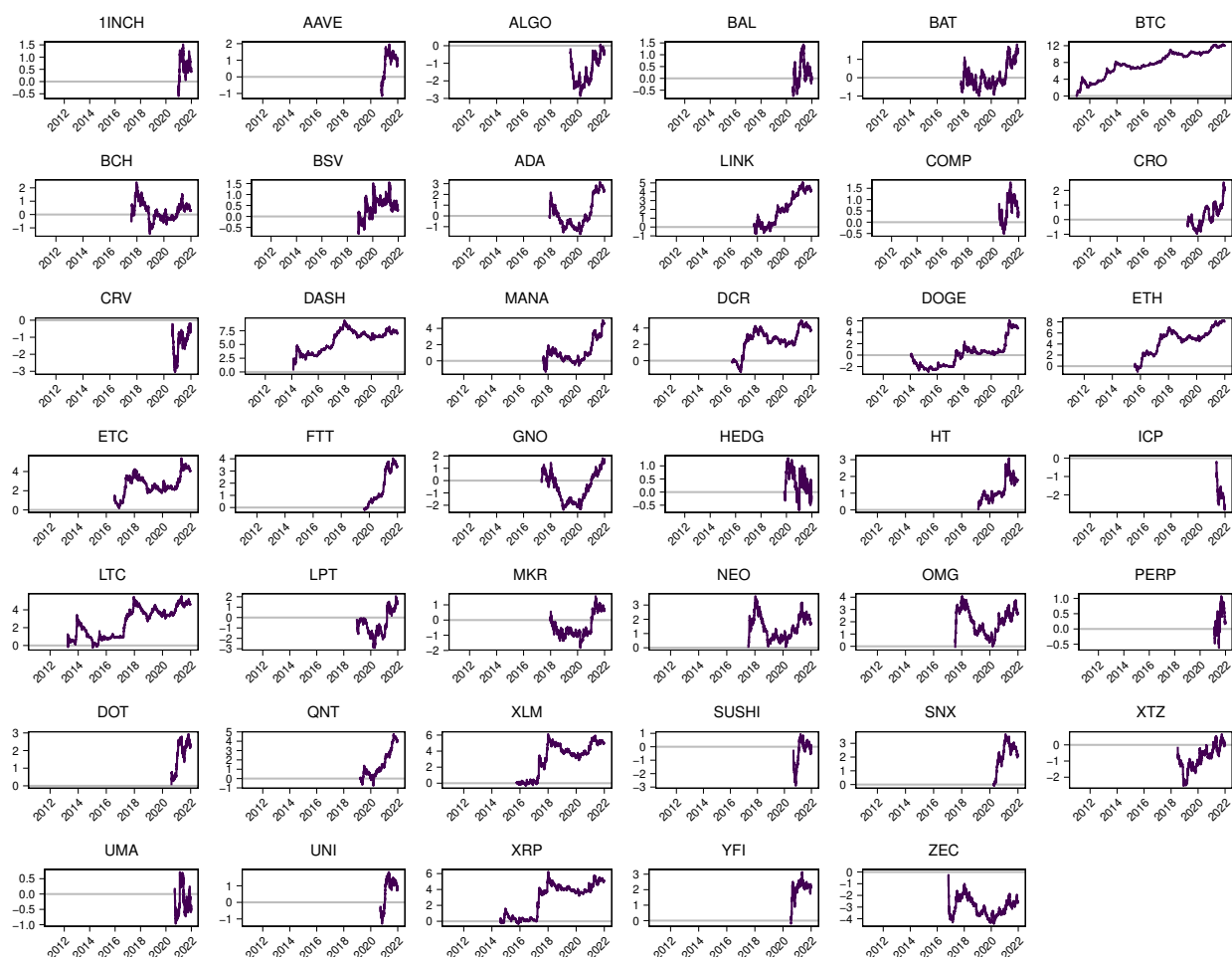


Figure 1. Log cumulative excess returns

Each panel depicts the log cumulative excess return for the cryptocurrency (denoted by its symbol) in the panel heading.

Figure 1 depicts log cumulative excess returns for the cryptocurrencies, with each cryp-

tocurrency identified by its symbol in the panel heading.⁶ In line with the volatilities and minimum and maximum values in Table 1, the cryptocurrencies display large swings in value in Figure 1. There is a tendency for the swings to correlate across cryptocurrencies; for example, numerous cryptocurrencies evince large appreciations followed by prolonged depreciations in 2017 to 2018, while many experience sizable appreciations during the advent of COVID-19 in 2020.

2.2. Predictors

We consider a broad array of predictors for forecasting daily cryptocurrency excess returns. The first set of predictors is based on data from CoinMetrics.⁷

Network value Supply of the cryptocurrency in US dollars at the end of that day (also known as “market capitalization”). It is computed using the daily closing price on that day.

Network value-to-transaction ratio Network value divided by the adjusted transfer value for that day. The adjusted transfer value is computed using the native units network value and adjusted transaction volume.

Active address-to-network value ratio Number of unique addresses that are active in the network (either as a destination or source of a ledger change) on that day divided by network value.

⁶The symbols for the 41 cryptocurrencies are as follows: 1inch (1INCH), Aave (AAVE), Algorand (ALGO), Balancer (BAL), Basic Attention Token (BAT), Bitcoin (BTC), Bitcoin Cash (BCH), Bitcoin SV (BSV), Cardano (ADA), Chainlink (LINK), Compound (COMP), Crypto.com Coin (CRO), Curve DAO Token (CRV), Dash (DASH), Decentraland (MANA), Decred (DCR), Dogecoin (DOGE), Ethereum (ETH), Ethereum Classic (ETC), FTX Token (FTT), Gnosis (GNO), HedgeTrade (HEDG), Huobi Token (HT), Internet Computer (ICP), Litecoin (LTC), Livepeer (LPT), Maker (MKR), Neo (NEO), OMGNetwork (OMG), Perpetual Protocol (PERP), Polkadot (DOT), Quant (QNT), Stellar (XLM), SushiSwap (SUSHI), Synthetix (SNX), Tezos (XTZ), UMA (UMA), Uniswap (UNI), XRP (XRP), yearn.finance (YFI), Zcash (ZEC).

⁷See the documentation on the [CoinMetrics](https://coinmetrics.io/) website for more information on the predictors.

Address-to-network value ratio Number of unique addresses holding any amount of native units at the end of that day divided by network value.

Velocity Value transferred over the last year divided by the current supply on that day; a measure of turnover.

Number of transactions Number of transactions on that day. Transactions are counted whether or not they involve the transfer of native units from one ledger entity to another.

Number of transfers Number of transfers on that day. Transfers constitute movements of native units from one ledger entity to another.

Mean dollar size of transfers US dollar value of native units transferred divided by the number of transfers on that day.

Median dollar size of transfers Median US dollar value transferred per transfer on that day.

Active supply Number of unique native units that transacted at least once in the last 30 days.

Realized network value US dollar value of the supply of the cryptocurrency at the end of that day, based on the closing price on the day that a native unit last moved.

Current-to-realized ratio Network value divided by realized network value.

Supply equality ratio Supply of the cryptocurrency held by all addresses with less than one-ten-millionth of the current supply of native units divided by the supply held by the top one percent of addresses on that day.

Volatility Standard deviation of the log return over the last 30 days.

We use cumulative returns to measure time-series momentum (Moskowitz et al., 2012) and prices to compute technical signals based on moving averages, which are popular with trend-following traders.

7-day time-series momentum 7-day cumulative excess return.

28-day time-series momentum 28-day cumulative excess return.

Price signal Price-based moving-average signal.

We also construct predictors designed to measure attention and sentiment via online activity and publications, beginning with Google Trends search volume.

Google Trends search volume Search-volume index for Google Trends searches involving the cryptocurrency's name or symbol for that day.

For a given query, the Google Trends search-volume index is computed based on the number of searches involving a term divided by the total number of searchers in a geographical region for a specific time period. The search-volume index provided by Google Trends is an integer between zero and 100, where the scale is determined using the smallest and largest values of the search ratios for the query's time period. Because Google Trends places limits on the amount of data that can be downloaded for a given query, we need to download the search-volume index data in batches. Due to the scaling of the search-volume index by Google Trends, figures for the index are not comparable over time for different batches. To address this issue, we download daily data in batches from January to July and July to January and use data from the overlapping months to compute adjustment factors for splicing together series that are comparable over time.

In addition, we download Reddit comments that appear in the "Cryptocurrency" subreddit and include the cryptocurrency's name or symbol. We combine the text for all of the comments for that day.

Number of Reddit comments Number of Reddit comments for that day.

Reddit sentiment score Number of positive words minus the number of negative words, all divided by the total number of words, for Reddit comments for that day. The positive and negative words are from the respective lists in [Loughran and McDonald \(2011\)](#).

Reddit uncertainty score Number of uncertainty words divided by the total number of words for Reddit comments for that day. The uncertainty words are from the list in [Loughran and McDonald \(2011\)](#).

Following standard practice, before calculating the sentiment and uncertainty scores, we eliminate stop words (e.g., “a,” “I,” “and,” “or”). We also tokenize the text by applying the [Porter \(1980\)](#) stemmer, which eliminates suffixes (e.g., “-tion”) from words to ensure that relevant words match those in the [Loughran and McDonald \(2011\)](#) lists.

Finally, we compute metrics based on articles from Factiva.⁸ Specifically, we download articles from the “Cryptocurrency” category that contain the cryptocurrency’s name or symbol. We combine the text for all of the articles for that day.

Number of Factiva articles Number of Factiva articles for that day.

Factiva article sentiment score Number of positive words minus the number of negative words, all divided by the total number of words, for words appearing in the bodies of Factiva articles for that day.

⁸Articles from Factiva are drawn from the following 47 publications from around the world: *The Coin-telegraph*, *CoinDesk.com*, *Blockonomi*, *Dow Jones Newswires*, *express.co.uk* (UK), *PR Newswire*, *CE NoticiasFinancieras* (Latin America), *Investing.com*, *Financial Times*, *Reuters*, *iCrowdNewswire*, *The Wall Street Journal*, *M2 Presswire*, *The Independent*, *Blockchain.News*, *The Times* (UK), *Investor’s Business Daily* (US), *The Telegraph* (UK), *MarketWatch*, *Brave New Coin*, *Sputnik News Service* (Russia), *Benzinga.com*, *Mondaq Business Briefing*, *Business Insider*, *CNN*, *Forbes*, *Business Wire*, *City AM* (London), *South China Morning Post*, *GlobeNewswire* (US), *Investment Weekly News*, *The Economic Times*, *ACCESSWIRE*, *Postmedia Breaking News* (Canada), *Hedge Week*, *Daily Mail*, *The Australian*, *Financial News* (Europe), *Exchange News Direct*, *Korea Times* (South Korea), *The Globe and Mail*, *Agence France Presse*, *Institutional Asset Manager*, *The Canadian Press*, *Barron’s*, *Times of India*, *The New York Times*.

Factiva article uncertainty score Number of uncertainty words divided by the total number of words for words appearing in the bodies of Factiva articles for that day.

Factiva headline sentiment score Number of positive words minus the number of negative words, all divided by the total number of words, for words appearing in the headlines of Factiva articles for that day.

Factiva headline uncertainty score Number of uncertainty words divided by the total number of words for words appearing in the headlines of Factiva articles for that day.

Table 2 lists the predictors, along with their abbreviations and transformations. We transform each predictor by computing log deviations (or, in some cases, simple deviations) from 7- and 28-day moving averages. This “feature engineering” helps to render the predictors stationary and provides sharper signals for forecasting cryptocurrency returns.⁹ We also include the 7- and 28-day time-series momentum in their levels (i.e., without transformation). After all of the transformations, we have a total of 54 predictors.

To aid in the interpretation of fitted models in Section 6, Table 2 divides the predictors into four groups. The first group is “Network,” which includes network value and valuation ratios relating to network value. Network value corresponds to the well-known size effect in the equity market (e.g., Banz, 1981), while the valuation ratios can be viewed as cryptocurrency counterparts to popular equity valuation ratios (Liu et al., 2023), such as dividend-price and book-to-market ratios. The second group is “Transactions” and is comprised of a variety of variables relating to activity on a cryptocurrency’s network. The first two groups together can be viewed as constituting “fundamentals” for forecasting cryptocurrency returns. “Market” is the next group and is made up of return volatility, time-series momentum, and price-based technical signals in the form of moving-average rules. Liu and Tsyvinski (2021) report in-sample evidence of time-series momentum in cryptocurrency re-

⁹The results are robust to the lengths of the moving averages; for example, log (or simple) deviations from 7- and 70-day moving averages produce similar results.

Table 2. Predictors

The table lists the predictors used to forecast daily cryptocurrency excess returns. The predictors are defined in the text. The third column provides the transformations applied to the predictors; MAs = moving averages.

(1) Predictor	(2) Abbreviation	(3) Transformations
Panel A: Network		
Network value	NV	Log deviations from 7- and 28-day MAs
Network value-to-transaction ratio	NVT	Log deviations from 7- and 28-day MAs
Active address-to-network value ratio	AdrActNV	Log deviations from 7- and 28-day MAs
Address-to-network value ratio	AdrNV	Log deviations from 7- and 28-day MAs
Panel B: Transactions		
Velocity	Vel1y	Log deviations from 7- and 28-day MAs
Number of transactions	NumTrx	Log deviations from 7- and 28-day MAs
Number of transfers	NumTrf	Log deviations from 7- and 28-day MAs
Mean dollar size of transfers	TrfSizeMean	Log deviations from 7- and 28-day MAs
Median dollar size of transfers	TrfSizeMed	Log deviations from 7- and 28-day MAs
Active supply	Sup30d	Log deviations from 7- and 28-day MAs
Realized network value	RealNV	Log deviations from 7- and 28-day MAs
Current-to-realized ratio	CRNV	Log deviations from 7- and 28-day MAs
Supply equality ratio	SER	Log deviations from 7- and 28-day MAs
Panel C: Market		
Volatility	Vol30d	Deviations from 7- and 28-day MAs
7-day time-series momentum	TSM7	Level, deviations from 7- and 28-day MAs
28-day time-series momentum	TSM28	Level, deviations from 7- and 28-day MAs
Price signal	PrcMA	Log deviations from 7- and 28-day MAs

turns, while [Huang et al. \(2019\)](#), [Detzel et al. \(2021\)](#), and [Gradojevic et al. \(2023\)](#) find that technical indicators are useful for forecasting Bitcoin returns. Finally, the “Online Activity” group of predictors includes Google Trends search volume and Reddit- and Factiva-based metrics. The variables in the last group reflect investor attention and sentiment surrounding a cryptocurrency.

Table 2 (continued)

(1) Predictor	(2) Abbreviation	(3) Transformations
Panel D: Online Activity		
Google Trends search volume	GTSrch	Log deviations from 7- and 28-day MAs
Number of Reddit comments	RedNumCom	Log deviations from 7- and 28-day MAs
Reddit sentiment score	RedSent	Log deviations from 7- and 28-day MAs
Reddit uncertainty score	RedUnc	Log deviations from 7- and 28-day MAs
Number of Factiva articles	FacNumArt	Log deviations from 7- and 28-day MAs
Factiva article sentiment score	FacSentArt	Deviations from 7- and 28-day MAs
Factiva article uncertainty score	FacUncArt	Deviations from 7- and 28-day MAs
Factiva headline sentiment score	FacSentHdl	Deviations from 7- and 28-day MAs
Factiva headline uncertainty score	FacUncHdl	Deviations from 7- and 28-day MAs

3. Forecasting Methods

We begin with the following general model for predicting the one-day-ahead cryptocurrency excess return:

$$r_{i,t+1} = f(\mathbf{x}_{i,t}; \boldsymbol{\eta}) + \varepsilon_{i,t+1}, \quad (1)$$

where $r_{i,t}$ is the day- t excess return for cryptocurrency i , $\mathbf{x}_{i,t} = [x_{1,i,t} \ \cdots \ x_{k,i,t}]'$ is a k -vector of predictors, $f(\mathbf{x}_{i,t}; \boldsymbol{\eta})$ is the conditional expectation (or prediction) function that depends on a vector of parameters $\boldsymbol{\eta}$, and $\varepsilon_{i,t+1}$ is a zero-mean and serially uncorrelated disturbance term. Observe that Equation (1) is a pooled model, as it assumes that the prediction function and its parameters are the same across cryptocurrencies. The parameter homogeneity assumption entailed by pooling substantially reduces the number of parameters that we need to estimate—since we do not have to estimate a separate set of parameters for each cryptocurrency—which helps to improve out-of-sample performance in light of the bias-variance trade-off. This is particularly helpful for cryptocurrencies that were introduced more recently and thus have relatively few observations for fitting the machine-learning models.

In the context of asset return prediction using machine learning, [Freyberger et al. \(2020\)](#) and [Gu et al. \(2020\)](#), among others, use a pooled approach to forecast individual stock returns, while [Filippou et al. \(2023\)](#) employ such an approach to forecast exchange rate changes.

Based on Equation (1), the excess return forecast is given by

$$\hat{r}_{i,t+1} = \hat{f}(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}), \quad (2)$$

where $\hat{f}(\cdot; \hat{\boldsymbol{\eta}})$ is the fitted prediction function based on data through day t . By fitting the prediction function using data through day t , we ensure that there is no look-ahead bias in the forecast. We consider a variety of machine-learning methods for estimating the prediction function.

3.1. Linear-ENet

We first consider a linear specification for the prediction function, where we fit the linear model via the ENet ([Zou and Hastie, 2005](#)). The linear-ENet excess return forecast (abstracting from the intercept term for notational simplicity) can be expressed as

$$\hat{r}_{i,t+1}^{\text{LinENet}} = \mathbf{x}'_{i,t} \hat{\boldsymbol{\eta}}, \quad (3)$$

where $\hat{\boldsymbol{\eta}}$ is the vector of coefficients for the linear model estimated via the ENet based on data through t . By construction, conventional OLS estimation of a linear model maximizes the fit of the model over the training sample. However, especially in the presence of a large number of correlated predictors and a low signal-to-noise ratio, OLS estimation is highly susceptible to overfitting the model to the training data, which harms out-of-sample performance. The ENet is an extension of the seminal LASSO ([Tibshirani, 1996](#)). Like the LASSO, the ENet uses penalized (or regularized) regression to shrink the parameter estimates toward zero, thereby helping to guard against overfitting.

The ENet objective function for the linear model can be expressed as

$$\arg \min_{\boldsymbol{\eta}} \frac{1}{2(t-1)n} \left[\sum_{i=1}^n \sum_{s=1}^{t-1} (r_{i,s+1} - \mathbf{x}'_{i,s} \boldsymbol{\eta})^2 \right] + \lambda P_{\delta}(\boldsymbol{\eta}), \quad (4)$$

where

$$P_{\delta}(\boldsymbol{\eta}) = 0.5(1 - \delta) \|\boldsymbol{\eta}\|_2^2 + \delta \|\boldsymbol{\eta}\|_1; \quad (5)$$

$\lambda \geq 0$ is a hyperparameter that governs the degree of shrinkage; $\|\cdot\|_1$ and $\|\cdot\|_2$ are the ℓ_1 and ℓ_2 norms, respectively; $0 \leq \delta \leq 1$ is a hyperparameter for blending the ℓ_1 and ℓ_2 components in the penalty term; and n is the number of cryptocurrencies.¹⁰ The ENet objective function in Equation (4) reduces to that for OLS when $\lambda = 0$. If $\delta = 1$, then Equation (4) corresponds to the LASSO objective function, while $\delta = 0$ corresponds to ridge regression (Hoerl and Kennard, 1970). Because Equation (5) includes an ℓ_1 component (as in the LASSO), it permits shrinkage to exactly zero, so the ENet also performs variable selection. We tune the hyperparameters λ and δ using a walk-forward cross-validation strategy that respects the time-series dimension of the panel data (as described in Section 3.5).

3.2. Random Forest

The linear-ENet forecast in Equation (3) is based on a linear approximation to the prediction function for the general model in Equation (1). Next, we consider three machine-learning techniques that provide flexible nonlinear approximations to the prediction function, beginning with the random forest (Breiman, 2001). A random forest is based on decision trees, which allow for multiway interactions and higher-order effects of predictors. In essence, a decision tree partitions the predictor space into non-overlapping regions and assigns a prediction (or score) for the target in each region. The partitions are generated by a sequence of splitting rules, typically based on the classification and regression tree (CART) algorithm

¹⁰For notational simplicity, Equation (4) assumes a balanced panel. For our application, the panel is unbalanced. It is straightforward to modify the notation for an unbalanced panel.

(Breiman et al., 1984). The split at the top of a tree is the “root node,” while the final set of subgroups defining the predictive regions at the bottom of the tree is comprised of “terminal” or “leaf nodes”; the splits in between are “internal nodes.” For a regression problem, the prediction is the average value of the target observations in a given leaf node. The forecast corresponding to a regression tree with U leaf nodes can be expressed as

$$\hat{r}_{i,t+1}^{\text{RegTree}} = \sum_{u=1}^U \bar{r}_u \mathbf{1}_u(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_u), \quad (6)$$

where the indicator function $\mathbf{1}_u(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_u) = 1$ if $\mathbf{x}_{i,t} \in R_u(\hat{\boldsymbol{\eta}}_u)$ for the u th region denoted by R_u (corresponding to the parameter vector $\hat{\boldsymbol{\eta}}_u$ that is based on the splits defining the tree) and zero otherwise, and $\bar{r}_u$ is the average value of the target observations in R_u for the training sample.

A regression tree with a large number of leaves (i.e., a “deep” tree) can capture complex nonlinear predictive relations in the data.¹¹ Although a deep tree substantially reduces the bias of the fitted tree, its high variance makes it susceptible to overfitting. To reduce the variance and thereby improve out-of-sample performance in light of the bias-variance trade-off, a random forest employs bagging (Breiman, 1996) by averaging forecasts over many deep trees, where each tree is constructed based on a bootstrap sample of the original training data. To further reduce the variance by “decorrelating” the trees when executing the bootstrap, each split is based on a randomly selected subset of the predictors. Indexing the bootstrap sample by b , the random forest forecast is given by

$$\hat{r}_{i,t+1}^{\text{RanFor}} = \frac{1}{B} \sum_{b=1}^B \left[\sum_{u=1}^U \bar{r}_u^{(b)} \mathbf{1}_u^{(b)}(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_u) \right], \quad (7)$$

where B is the number of bootstrap samples, and $\bar{r}_u^{(b)}$ and $\mathbf{1}_u^{(b)}(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_u)$ are the analogs to $\bar{r}_u$ and $\mathbf{1}_u(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_u)$, respectively, in Equation (6) for the b th bootstrap sample. We tune the following hyperparameters for the random forest via walk-forward cross-validation: number

¹¹In the extreme, if the tree is grown so that there is only one target observation in each leaf node, then the regression tree is “fully grown.” A fully grown tree perfectly fits (or interpolates) the training data.

of trees in the forest (B), maximum depth of each tree, minimum number of observations needed to split an internal node, minimum number of observations needed to be at a leaf node, number of predictors to consider for a split, and impurity threshold for splitting a node.¹²

3.3. XGBoost

A boosted tree is another approach for forecasting with a regression tree. It is based on gradient boosting (Breiman, 1997; Friedman, 2001), a sequential ensemble method for improving out-of-sample prediction. The idea is to fit the prediction function additively:

$$\hat{f}(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}) = \sum_{m=1}^M \hat{f}_m(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_m). \quad (8)$$

Each function $\hat{f}_m(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_m)$ on the right-hand-side of Equation (8) is a “weak” learner (i.e., a relatively simple model). Although relatively simple models help to guard against overfitting, they are more likely to suffer from a sizable bias and thus evince poor fit. Boosting improves the fit by adding another $\hat{f}_m(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_m)$ element to model the residuals from the previous function in the sequence. In this way, boosting refines a sequence of simple models to reduce the bias. In the context of decision trees, boosting entails constructing a sequence of relatively “shallow” trees, which are then combined into an ensemble. Random forests and boosted trees follow different tacks to improve out-of-sample performance: a random forest begins with a deep tree with low bias and uses bagging across a large number of trees to reduce the variance; a boosted tree begins with a shallow tree with low variance and refines the tree to reduce the bias.

To make boosting more robust, Friedman (2002) proposes stochastic gradient boosting. In the spirit of bagging, instead of basing each $\hat{f}_m(\mathbf{x}_{i,t}; \hat{\boldsymbol{\eta}}_m)$ in the sequence on all of the training data, each element is based on a randomly drawn (without replacement) subsam-

¹²The [sklearn.ensemble.RandomForestRegressor](https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html) documentation provides details for the hyperparameters for the random forest.

ple of the data. We fit boosted trees via stochastic gradient boosting using the XGBoost algorithm (Chen and Guestrin, 2016). We tune the following hyperparameters for the XGBoost algorithm using walk-forward cross-validation: number of trees in the sequence (M), step-size shrinkage for gradient boosting, minimum loss reduction needed to make a split, maximum depth of a tree, minimum number of observations in a child after splitting, subsample ratio for the training data for each tree, subsample ratio of predictors for each tree, and ℓ_1 and ℓ_2 regularization hyperparameters.¹³

3.4. Deep Neural Network

A feedforward neural network is another machine-learning technique that provides a flexible nonlinear approximation to the prediction function. A neural network is comprised of multiple layers. The set of predictors makes up the first (or “input”) layer. Next, there are $L \geq 1$ “hidden” layers, where each hidden layer l contains P_l neurons. Each neuron takes signals from the neurons in the previous layer to generate a new signal:

$$h_m^{(l)} = g\left(\omega_{m,0}^{(l)} + \sum_{j=1}^{P_{l-1}} \omega_{m,j}^{(l)} h_j^{(l-1)}\right) \quad \text{for } m = 1, \dots, P_l; l = 1, \dots, L, \quad (9)$$

where $h_m^{(l)}$ is the signal corresponding to the m th neuron in the l th hidden layer¹⁴, $\omega_{m,0}^{(l)}, \omega_{m,1}^{(l)}, \dots, \omega_{m,P_{l-1}}^{(l)}$ are weights; and $g(\cdot)$ is a nonlinear activation function. The final layer is the “output” layer that takes the signals from the last hidden layer and transforms them into a prediction:

$$\hat{r}_{i,t+1}^{\text{Net}} = \omega_0^{(L+1)} + \sum_{j=1}^{P_L} \omega_j^{(L+1)} h_j^{(L)}. \quad (10)$$

¹³The XGBoost documentation provides details for the XGBoost algorithm, including its hyperparameters.

¹⁴For the first hidden layer, $h_j^{(0)} = x_{j,i,t}$ for $j = 1, \dots, k$.

For the activation function, we use the popular rectified linear unit (ReLU) or “leaky” ReLU (Maas et al., 2013).¹⁵

The network interactions and activation function allow for complex nonlinearities as the inputs feed through to the hidden layers and finally to the output layer. Although a single hidden layer with enough neurons is theoretically sufficient to approximate any smooth function (e.g., Cybenko, 1989; Funahashi, 1989; Hornik et al., 1989; Hornik, 1991), neural networks with multiple hidden layers are commonly used, as there are performance advantages to including multiple hidden layers (e.g., Goodfellow et al., 2016; Rolnick and Tegmark, 2018). Accordingly, we consider a “deep” neural network with three hidden layers.

Fitting a neural network entails estimating the weights, which is usually done via a stochastic gradient descent (SGD) algorithm. We fit the deep neural network by minimizing the training sample MSE using the Adam SGD algorithm (Kingma and Ba, 2015). We use walk-forward cross-validation to tune the following: activation function (ReLU or leaky ReLU); dropout rate (Srivastava et al., 2014); learning rate for SGD; and the number of neurons in the first, second, and third hidden layers (up to 100, 50, and 30, respectively).¹⁶

3.5. Hyperparameter Tuning

We tune the hyperparameters for the machine-learning models via walk-forward five-fold cross-validation. The “walk-forward” aspect of the procedure respects the time-series dimension of our panel data. We divide the available data at the time of forecast formation into initial training and validation samples, where the latter is comprised of panel data observations for the last 300 days. The validation sample is further divided into five folds comprised of the panel data observations for the first through fifth 60-day periods of the

¹⁵The ReLU function is $g(x) = \max\{x, 0\}$. For the leaky ReLU, $g(x) = x$ if $x > 0$ and $0.01x$ if $x \leq 0$. The leaky ReLU adjusts the conventional ReLU to help prevent the neural network from “dying,” meaning that no neurons are activated in the network.

¹⁶For the Adam SGD algorithm, we allow up to 100 epochs (with early stopping based on a validation loss) and use a batch size of 128. To reduce the influence of weight initialization in the algorithm, we fit the model five times and take an average of the forecasts generated by the five fitted models.

validation sample.

For a given vector of hyperparameter values, we fit the model using the initial training sample, generate forecasts for the first fold of the validation sample for the available cryptocurrencies, and store the MSE. Next, we fit the model using panel data observations for the initial training sample and the first fold of the validation sample, generate forecasts for the second fold of the validation sample for the available cryptocurrencies, and store the MSE. We proceed in this manner through the remaining folds of the validation sample and compute the average MSE over the five folds. Finally, we select the vector of hyperparameters that minimizes the average MSE.¹⁷

We typically consider a lengthy grid of values for each of the hyperparameters. For the random forest, XGBoost, and deep neural network, it becomes computationally infeasible to compute the walk-forward cross-validation MSE for all possible combinations of hyperparameter values implied by the grids. Instead, we use the **Optuna** (Akiba et al., 2019) framework, which employs Bayesian optimization to smartly (instead of exhaustively) search the hyperparameter space to select a vector of hyperparameter values that is likely to be optimal or nearly so. After tuning the hyperparameters, we train the model using all of the available data at the time of forecast formation to generate the forecasts (i.e., we use an expanding estimation window). We tune the hyperparameters and train the models every 30 days using data for the available cryptocurrencies at the time of forecast formation.

3.6. Ensembles

We also consider a pair of ensemble (or combination) forecasts:

Ensemble-nonlinear Average of the random forest, XGBoost, and neural network forecasts.

Ensemble-all Average of the linear-ENet, random forest, XGBoost, and neural network

¹⁷The results are robust to other fold patterns (e.g., ten folds comprised of panel data observations for the first through tenth 30-day periods of the validation sample).

forecasts.

Ensembles often perform nearly as well as or better than the best individual forecast, and they are frequently used in machine-learning applications. Intuitively, averaging across multiple forecasts can reduce the risk of relying on a single forecast—similarly to diversifying across assets to reduce portfolio risk—which can improve out-of-sample performance (Timmermann, 2006; Rapach et al., 2010). Ensembles are also useful in a practical sense in that it is difficult to know a priori the best individual forecast, and they accommodate potential complementarities between the individual forecasts.

4. Forecasting Accuracy

This section analyzes the statistic accuracy of the daily cryptocurrency excess return forecasts based on machine-learning methods.

4.1. Accuracy Metrics

We compare cryptocurrency excess return forecasts based on the predictors in Section 2.2 and the machine-learning methods in Section 3 to the prevailing mean benchmark forecast. The prevailing mean benchmark for cryptocurrency i is the average of the excess return observations available at the time of forecast formation:

$$\hat{r}_{i,t+1}^{\text{Bench}} = \left(\frac{1}{t}\right) \sum_{s=1}^t r_{i,s}. \quad (11)$$

The prevailing mean forecast in Equation (11) corresponds to the constant expected excess return model:

$$r_{i,t+1} = \mu_i + \varepsilon_{i,t+1}, \quad (12)$$

where μ_i is the unconditional mean. Equation (12) assumes that the excess return is unpredictable (apart from its unconditional mean). Because asset returns contain an inherently large unpredictable component, the prevailing mean benchmark is a relevant and stringent

benchmark for assessing return predictability (e.g., [Goyal and Welch, 2008](#)).¹⁸

We can conveniently compare the out-of-sample MSE for the prevailing mean benchmark to that of a competing forecast via the out-of-sample R^2 statistic ([Fama and French, 1989](#); [Campbell and Thompson, 2008](#)):

$$R_{i,OS}^2 = 1 - \frac{\sum_{s=1}^{T-t_{in}} \left(r_{i,t_{in}+s} - \hat{r}_{i,t_{in}+s}^{Compete} \right)^2}{\sum_{s=1}^{T-t_{in}} \left(r_{i,t_{in}+s} - \hat{r}_{i,t_{in}+s}^{Bench} \right)^2}, \quad (13)$$

where $\hat{r}_{i,t}^{Compete}$ generically denotes a competing forecast, and t_{in} (T) is the end of the initial in-sample period (total sample). Equation (13) measures the proportional reduction in out-of-sample MSE for the competing forecast vis-à-vis the prevailing mean benchmark. We use the [Diebold and Mariano \(1995\)](#) and [West \(1996\)](#) (DMW) statistic to test whether the competing forecast delivers a statistically significant reduction in MSE relative to the prevailing mean benchmark. The DMW statistic can be computed as the t -statistic corresponding to the intercept a_i in the following time-series regression:

$$\underbrace{\left(r_{i,t} - \hat{r}_{i,t}^{Bench} \right)^2 - \left(r_{i,t} - \hat{r}_{i,t}^{Compete} \right)^2}_{d_{i,t}} = a_i + \varepsilon_{i,t} \quad \text{for } t = t_{in} + 1, \dots, T, \quad (14)$$

where $d_{i,t}$ is the day- t loss differential (i.e., the difference between the squared errors for the benchmark and competing forecasts). We test the null hypothesis $H_0: a_i \leq 0$ against the (one-sided, upper-tail) alternative $H_A: a_i > 0$, which is tantamount to testing $H_0: R_{i,OS}^2 \leq 0$ against $H_A: R_{i,OS}^2 > 0$.

We also compute a pooled version of Equation (13) for the entire set of cryptocurrencies taken together:

$$R_{All,OS}^2 = 1 - \frac{\sum_{i=1}^n \sum_{s=1}^{T-t_{in}} \left(r_{i,t_{in}+s} - \hat{r}_{i,t_{in}+s}^{Compete} \right)^2}{\sum_{i=1}^n \sum_{s=1}^{T-t_{in}} \left(r_{i,t_{in}+s} - \hat{r}_{i,t_{in}+s}^{Bench} \right)^2}. \quad (15)$$

¹⁸We use the first 28 days of available excess return observations as an initial in-sample estimation period for computing the prevailing mean forecast.

Equation (15) is the proportional reduction in out-of-sample MSE for the competing forecast vis-à-vis the benchmark forecast across all of the cryptocurrencies. In the context of Equation (15), we compute the DMW statistic via the t -statistic corresponding to the intercept a in the following pooled regression:

$$d_{i,t} = a + \varepsilon_{i,t} \quad \text{for } i = 1, \dots, n; t = t_{\text{in}} + 1, \dots, T. \quad (16)$$

We test $H_0: a \leq 0$ ($R_{\text{All,OS}}^2 \leq 0$) against $H_A: a > 0$ ($R_{\text{All,OS}}^2 > 0$). When computing the DMW statistic using Equation (16), we account for cross-sectional dependency by clustering the standard error by cryptocurrencies.

4.2. Out-of-Sample Results

Table 3 reports $R_{i,\text{OS}}^2$ statistics for daily excess return forecasts for each cryptocurrency and the different machine-learning methods. The start of the out-of-sample period for each cryptocurrency is provided in the second column. We begin computing out-of-sample forecasts on January 1, 2014 for the two available cryptocurrencies at that time (Bitcoin and Litecoin). This corresponds with the 2014 starting date used by Liu et al. (2022) to compute cryptocurrency factors as long-short portfolios based on sorted cryptocurrency characteristics. When a cryptocurrency becomes available, we include it in the panel data and begin computing out-of-sample forecasts for the cryptocurrency after it has been in the panel for 28 days.

Nearly all of the $R_{i,\text{OS}}^2$ statistics are positive in Table 3, so the different forecasting methods consistently outperform the prevailing mean benchmark in terms of out-of-sample MSE.¹⁹ For the linear-ENet forecast, 39 of the 41 $R_{i,\text{OS}}^2$ statistics are positive, and 18 are significant at conventional levels according to the DMW statistic. Turning to the decision trees, the random forest and XGBoost forecasts both have a lower MSE than the prevailing

¹⁹In contrast, as anticipated (see Section 3.1), a linear forecast based on conventional OLS estimation substantially underperforms the prevailing mean benchmark for all of the cryptocurrencies.

Table 3. Out-of-sample R^2 statistics

The table reports out-of-sample R^2 statistics in percent for daily excess return forecasts for the cryptocurrency in the first column. The start of the out-of-sample period for forecast evaluation is given in the second column; the end of the out-of-sample period is 2021-12-31 for all of the cryptocurrencies. The “All” row reports out-of-sample R^2 statistics for all 41 of the cryptocurrencies taken together. The out-of-sample R^2 statistic measures the proportional reduction in out-of-sample mean squared error (MSE) for the competing forecast in the column heading vis-à-vis the prevailing mean benchmark forecast; based on the Diebold and Mariano (1995) and West (1996) test, *, **, and *** indicate that the reduction in out-of-sample MSE is significant at the 10%, 5%, and 1% level, respectively. The penultimate (last) row reports the number of positive (significant at the 10% level) out-of-sample R^2 statistics for the 41 cryptocurrencies.

(1) Cryptocurrency	(2) Out-of-sample start	(3) Linear- ENet	(4) Random forest	(5) XGBoost	(6) Neural network	(7) Ensemble- nonlinear	(8) Ensemble- all
1inch	2021-01-24	1.31	1.53	1.72	1.75	1.98	1.95
Aave	2020-11-08	1.55**	3.01	3.71*	1.75**	3.04*	2.73**
Algorand	2019-07-21	1.27	1.53*	1.56	1.35	1.59*	1.53*
Balancer	2020-07-24	0.94**	2.26**	2.21**	1.35**	2.10**	1.88**
Basic Attention Token	2017-11-04	0.48*	1.73**	2.11**	1.07***	1.93***	1.64***
Bitcoin	2014-01-01	-0.05	0.22	1.98***	1.06**	1.58***	1.31***
Bitcoin Cash	2017-08-30	0.48	-0.78	-0.35	0.54	0.21	0.43
Bitcoin SV	2018-12-14	0.29*	0.50	0.78**	0.78***	0.81***	0.73***
Cardano	2017-12-30	2.47**	2.80**	4.45***	2.87**	3.64***	3.40**
Chainlink	2017-10-28	0.33	0.66	0.73	0.45	0.87*	0.81**
Compound	2020-07-17	0.40	0.95	1.40	0.75	1.25	1.14
Crypto.com Coin	2019-04-18	0.41	0.64*	1.52*	0.63**	1.06**	0.92**
Curve DAO Token	2020-09-13	1.40	0.12	0.39	1.66	0.97	1.19
Dash	2014-03-09	4.72***	4.77***	6.16***	6.32**	6.12***	5.88***
Decentraland	2017-09-23	1.12*	1.36*	1.55**	1.64**	1.61**	1.51**

mean benchmark for 37 cryptocurrencies, and 18 and 26, respectively, of the improvements in MSE are significant. For the neural network forecast, all 41 of the $R^2_{i,OS}$ statistics are positive, and 27 are significant. Taking all of the cryptocurrencies together (see the “All”

Table 3 (continued)

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cryptocurrency	Out-of-sample start	Linear- ENet	Random forest	XGBoost	Neural network	Ensemble- nonlinear	Ensemble- all
Decred	2016-06-15	0.18	0.07	1.29**	0.81*	0.94***	0.79***
Dogecoin	2014-02-21	0.19	0.36	1.33**	0.52**	0.87***	0.74**
Ethereum	2015-09-06	0.14	1.72**	2.40**	0.38	2.09***	1.72***
Ethereum Classic	2016-08-23	3.29***	4.02***	4.74***	3.52***	4.33***	4.12***
FTX Token	2019-09-18	0.37	2.32	1.77*	0.46*	1.71*	1.43*
Gnosis	2017-05-31	0.50*	0.68	1.04*	0.94***	1.07**	0.97**
HedgeTrade	2019-12-01	−0.01	0.14	0.76	0.17	0.51*	0.42*
Huobi Token	2019-04-04	0.45	−0.51	−0.19	0.97	0.54	0.71
Internet Computer	2021-06-09	4.12	3.42	3.95	3.66	3.86	4.00
Litecoin	2014-01-01	0.68**	0.77	2.11***	0.85*	1.55***	1.43***
Livepeer	2019-01-18	0.98**	1.60*	2.03**	1.21**	1.69**	1.55**
Maker	2018-01-24	0.13	0.41	1.41**	0.40*	0.94***	0.78**
Neo	2017-08-13	2.54***	3.35***	3.64***	3.31***	3.67***	3.48***
OMGNetwork	2017-08-13	5.15***	6.40***	6.22***	5.58***	6.31***	6.11***
Perpetual Protocol	2021-03-05	0.96***	0.16	−0.45	1.50***	1.03	1.19
Polkadot	2020-09-18	0.77	0.53	1.98	1.38	1.58	1.54
Quant	2019-04-14	0.32	1.25	1.80**	0.64*	1.34**	1.11**
Stellar	2015-10-29	0.17	−0.02	0.86***	0.05	0.45***	0.41***
SushiSwap	2020-09-30	2.43**	3.81**	4.04**	3.11**	3.82**	3.52**
Synthetic	2020-05-08	0.30	0.60	1.20	0.47	0.87	0.75

row in Table 3), the $R^2_{\text{All,OS}}$ statistics are 1.11%, 1.45%, 2.08%, and 1.54% for the linear-ENet, random forest, XGBoost, and neural network forecasts, respectively, all of which are significant at the 1% level.

According to the last two columns of Table 3, the ensembles provide effective strategies for forecasting daily cryptocurrency excess returns. The ensemble-nonlinear forecast—an average of the random forest, XGBoost, and neural network forecasts—generates positive $R^2_{i,\text{OS}}$

Table 3 (continued)

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cryptocurrency	Out-of-sample start	Linear- ENet	Random forest	XGBoost	Neural network	Ensemble- nonlinear	Ensemble- all
Tezos	2018-07-29	0.35	-0.30	-0.08	0.18	0.15	0.31
UMA	2020-10-07	1.54*	2.49*	1.38	1.60*	1.95*	1.91*
Uniswap	2020-10-17	1.82**	5.15*	4.21	2.81**	4.28*	3.76**
XRP	2014-09-13	0.29	0.64*	1.71***	0.85**	1.31***	1.11***
yearn.finance	2020-08-23	3.52	5.55*	6.27**	4.26*	5.49*	5.06*
Zcash	2016-11-27	1.00*	1.87**	2.11**	1.95***	2.23**	2.01**
All	2014-01-01	1.11***	1.45***	2.08***	1.54***	1.91***	1.78***
Number > 0		39	37	37	41	41	41
Number sig.		18	18	26	27	31	31

statistics for all of the cryptocurrencies, 31 of which are significant. Taking the cryptocurrencies together, the $R^2_{\text{All,OS}}$ statistic for the ensemble-nonlinear forecast is 1.91% (significant at the 1% level). Similarly to the ensemble-nonlinear forecast, the ensemble-all forecast outperforms the prevailing mean benchmark for all of the cryptocurrencies, and again 31 of the $R^2_{i,\text{OS}}$ statistics are significant. When we take all of the cryptocurrencies together, the ensemble-all produces an $R^2_{\text{All,OS}}$ statistic of 1.78% (significant at the 1% level).

Because daily asset returns inherently contain a large, unpredictable component, the $R^2_{i,\text{OS}}$ and $R^2_{\text{All,OS}}$ statistics will naturally be limited in magnitude. Nevertheless, they are often relatively sizable; furthermore, we show in Section 5 that the degree of out-of-sample excess return predictability indicated by the $R^2_{i,\text{OS}}$ and $R^2_{\text{All,OS}}$ statistics in Table 3 translates into substantial economic value. Many of the $R^2_{i,\text{OS}}$ statistics are above 1%, and they range from approximately 2.5% to well above 6% for cryptocurrencies such as Cardano, Dash, Ethereum Classic, Internet Computer, Neo, OMGNetwork, SushiSwap, and yearn.finance. These cryptocurrencies became available in 2017, 2014, 2016, 2021, 2017, 2017, 2020, and 2020, respectively, so it is not the case that only relatively young or old cryptocurrencies

evinced the strongest return predictability.

Overall, the results in Table 3 demonstrate that machine-learning techniques provide an effective means of extracting information from a wide range of predictors to improve daily cryptocurrency return prediction. All of the forecasts in Table 3 perform well, so the out-of-sample gains are robust across the different machine-learning methods as well as the ensembles.

While both the linear and nonlinear forecasts perform well in Table 3, the $R^2_{\text{All,OS}}$ statistics in the “All” row of Table 3 suggest that nonlinearities are important for improving out-of-sample performance. Specifically, the $R^2_{\text{All,OS}}$ statistic for the linear-ENet forecast (1.11%) is lower than those for the nonlinear random forest, XGBoost, and neural network forecasts (1.45%, 2.08%, and 1.54%, respectively) as well as the ensemble-nonlinear forecast (1.91%) that is an average of the three nonlinear forecasts. Thus, allowing for nonlinearities improves the accuracy of daily cryptocurrency return forecasts in terms of out-of-sample MSE.

We test for a significant difference between MSEs for the linear-ENet and a nonlinear forecast via a pooled DMW statistic, which corresponds to the t -statistic for the intercept a in a modified version of Equation (16):

$$(r_{i,t} - \hat{r}_{i,t}^{\text{LinENet}})^2 - (r_{i,t} - \hat{r}_{i,t}^{\text{Nonlin}})^2 = a + \varepsilon_{i,t} \quad \text{for } i = 1, \dots, n; t = t_{\text{in}} + 1, \dots, T, \quad (17)$$

where $\hat{r}_{i,t}^{\text{Nonlin}}$ generically denotes a nonlinear forecast. The left-hand side of Equation (17) is the loss differential between the linear-ENet and nonlinear forecasts. We continue to account for cross-sectional dependency by clustering the standard error by cryptocurrencies. The DMW statistics corresponding to Equation (17) are 3.69, 8.51, 5.29, and 9.83 for the random forest, XGBoost, neural network, and ensemble-nonlinear forecasts, respectively. We reject the null hypothesis in the upper tail at the 1% level for all four of the forecasts, so the nonlinear forecasts significantly reduce out-of-sample MSE compared to the linear forecast. We explore the role of nonlinearities further in Section 6.

Table 3 shows that daily cryptocurrency return forecasts based on a rich set of pre-

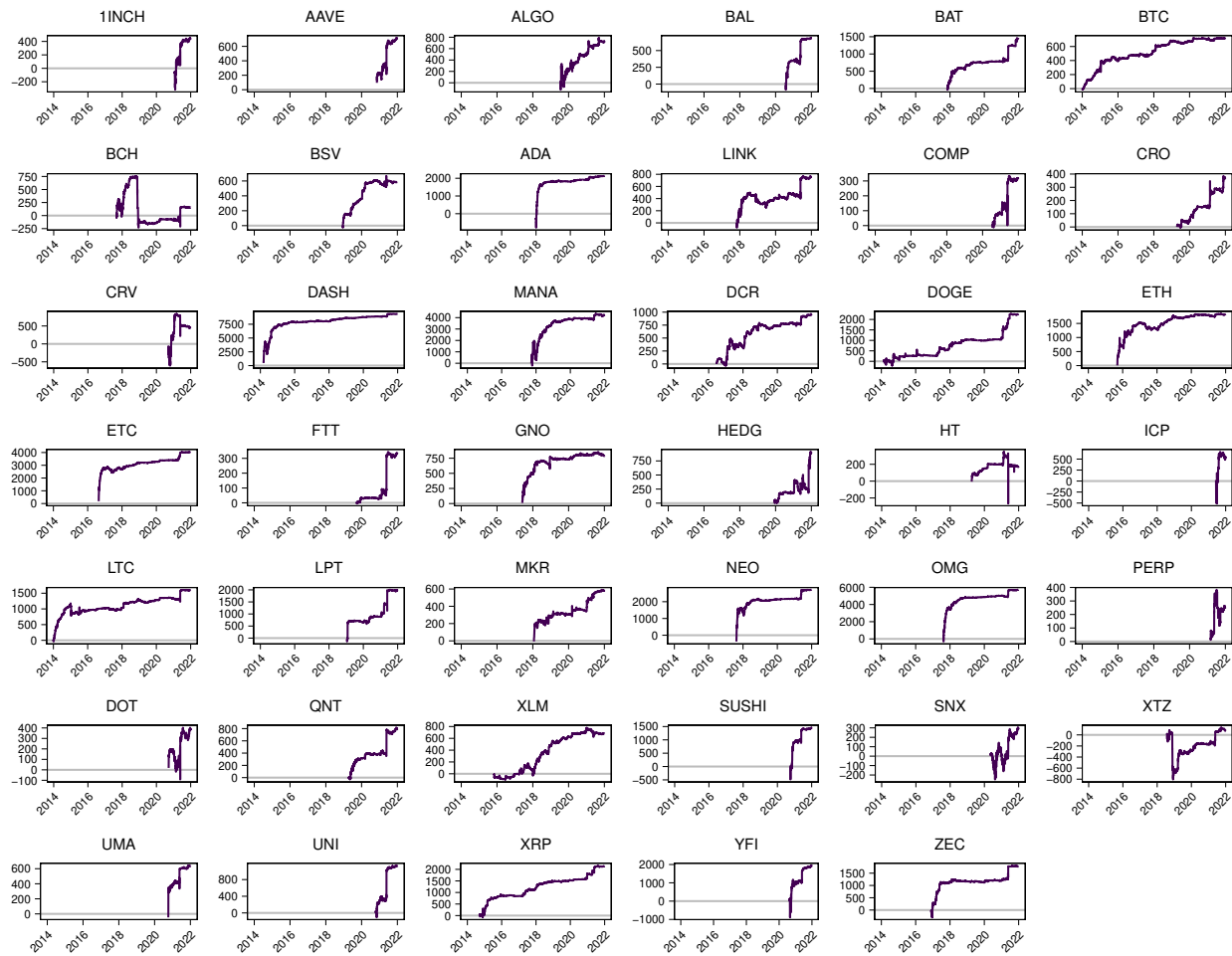


Figure 2. Cumulative differences in squared forecast errors

Each panel depicts the cumulative difference in squared forecast errors for the prevailing mean vis-à-vis the ensemble-nonlinear excess return forecast for the cryptocurrency (denoted by its symbol) in the panel heading.

dictors outperform the prevailing mean benchmark over the full forecast evaluation period available for each cryptocurrency. To get a sense of the consistency of the out-of-sample gains over time, Figure 2 depicts the cumulative difference in squared forecast errors for the prevailing mean benchmark vis-à-vis the ensemble-nonlinear forecast.²⁰ The cumulative

²⁰To conserve space, we focus on the ensemble-nonlinear forecast in Figure 2. As shown in Figures A1 to A5 in the Online Appendix, the plots for the other machine-learning forecasts are similar to those for the

difference in squared forecast errors is an informative graphical device suggested by Goyal and Welch (2003, 2008) for comparing the predictive accuracy of a competing forecast to that of a benchmark over time. We can determine if the competing forecast outperforms the benchmark in terms of out-of-sample MSE for any subsample by comparing the height of the curve at the beginning and end of the interval corresponding to the subsample. If the curve is higher (lower) at the end of the interval relative to the beginning, then the competing forecast outperforms (underperforms) the benchmark during the subsample. A uniformly positively sloped curve indicates that the competing forecast always outperforms the benchmark. While such an outcome is infeasible in practice with regard to daily asset return forecasting, more realistically, we seek a curve with a predominantly positive slope that avoids long segments with steeply negative slopes.

Figure 2 indicates that the ensemble-nonlinear forecast outperforms the prevailing mean benchmark consistently over time for the different cryptocurrencies. The slopes are positive the vast majority of the time, and there are many segments with steeply positive slopes, so the information in the predictors substantially improves out-of-sample accuracy. The curve is initially quite steeply positively sloped for some cryptocurrencies, such as Ethereum Classic (ETC), Neo (NEO), OMGNetwork (OMG), and UMA. This suggests that the information in the predictors is particularly useful for anticipating cryptocurrency returns for some cryptocurrencies when they are introduced as investors learn about their behavior. However, it is not the case that return predictability is always strongest at the outset of the out-of-sample period near the introduction of a cryptocurrency. For example, numerous cryptocurrencies that were introduced relatively early exhibit strong predictability from 2017 through 2018. Interestingly, this period coincides with the substantial appreciation and subsequent depreciation evident for many cryptocurrencies during this same period in Figure 1.

Figure 3 shows more clearly the strong performance of the ensemble-nonlinear forecast for the major swings in cryptocurrency values during 2017 and 2018, focusing on four of the ensemble-nonlinear forecast.

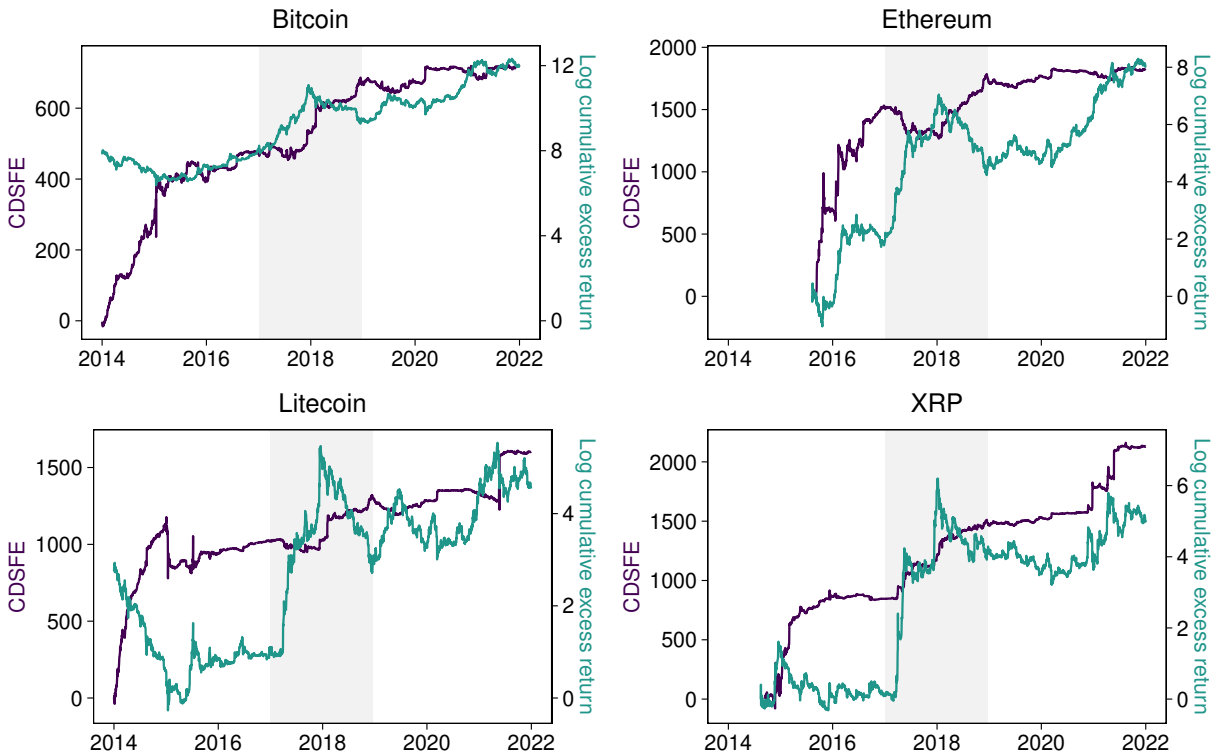


Figure 3. Results for four leading cryptocurrencies

Each panel depicts the log cumulative excess return in Figure 1 (right axis) and the cumulative difference in squared forecast errors (CDSFE) in Figure 2 (left axis) for Bitcoin, Ethereum, Litecoin, and XRP. The vertical bars depict 2017-01-01 to 2018-12-31.

leading cryptocurrencies: Bitcoin, Ethereum, Litecoin, and XRP.²¹ The figure superimposes the log cumulative excess returns from Figure 1 (right axis) on the cumulative differences in squared forecast errors from Figure 2 (left axis). Vertical bars indicate the beginning of 2017 through the end of 2018. For each cryptocurrency, the log cumulative excess return in Figure 3 exhibits a sharp appreciation during 2017 followed by a substantial depreciation in 2018. Throughout much of 2017 to 2018 (especially after early 2017), the plots of the cumulative differences in squared forecast errors are markedly positively sloped, highlighting

²¹As shown in Figures A6 to A10 in the Online Appendix, the plots for the ensemble-nonlinear forecast in Figure 3 are quite similar to those for the XGBoost, neural network, and ensemble-all forecasts; they are less similar to those for the linear-ENet and random forest forecasts.

the strong degree of return predictability during the major swings in the cryptocurrency values during 2017 to 2018.

Returning to Figure 2, numerous cryptocurrencies also have steeply positive slopes in early to mid 2020, corresponding to the substantive appreciations for many cryptocurrencies seen in Figure 1 during the advent of COVID-19. Overall, it appears that the information in the predictors becomes especially useful for predicting returns when cryptocurrencies experience large swings in value. Furthermore, reinforcing the consistency of the out-of-sample gains for many cryptocurrencies, the negatively sloped segments of the curves are typically short-lived and not very steep in Figure 2.

5. Economic Value

In this section, we examine the economic value of machine-learning forecasts of daily cryptocurrency excess returns in an investment context.

5.1. Investing in Individual Cryptocurrencies

We first consider a mean-variance investor who allocates between an individual cryptocurrency and risk-free Treasury bills. At the end of day t , the investor determines their allocations based on the following objective function:

$$\arg \max_{w_{i,t+1}} w_{i,t+1} \hat{r}_{i,t+1} - 0.5\gamma w_{i,t+1}^2 \hat{\sigma}_{i,t+1}^2, \quad (18)$$

where γ is the coefficient of relative risk aversion; $w_{i,t+1}$ and $1 - w_{i,t+1}$ are the day- $(t + 1)$ allocations to cryptocurrency i and risk-free bills, respectively, which are determined at the end of day t ; and $\hat{r}_{i,t+1}$ ($\hat{\sigma}_{i,t+1}^2$) generically denotes the investor's forecast of the cryptocurrency excess return (variance), which are based on data available through day t . The well-known solution to equation (18) is given by

$$w_{i,t+1}^* = \left(\frac{1}{\gamma} \right) \left(\frac{\hat{r}_{i,t+1}}{\hat{\sigma}_{i,t+1}^2} \right). \quad (19)$$

We assume that the investor uses an exponentially weighted moving-average estimator for $\hat{\sigma}_{i,t+1}^2$, which is popular among practitioners.²² To prevent the allocations from becoming implausible, we impose the restriction that $-1 \leq w_{i,t+1} \leq 2$. We assume that $\gamma = 3$; the results are similar for reasonable alternative values for γ .

We consider two cases. In the first, the investor uses the prevailing mean benchmark to forecast the cryptocurrency excess return in Equation (19), thereby ignoring return predictability; in the second, the investor instead uses a machine-learning forecast of the cryptocurrency excess return.²³ The average utility (or certainty equivalent return) realized by the investor is given by

$$\bar{U}_i^j = \bar{r}_i^j - 0.5\gamma\hat{\sigma}_i^{2,j} \quad \text{for } j = \text{Bench, Compete}, \quad (20)$$

where $\bar{r}_i^{\text{Bench}}$ ($\bar{r}_i^{\text{Compete}}$) and $\hat{\sigma}_i^{2,\text{Bench}}$ ($\hat{\sigma}_i^{2,\text{Compete}}$) are the mean and variance, respectively, for the portfolio excess return over the out-of-sample period when the investor uses the prevailing mean benchmark (competing machine-learning) forecast of the cryptocurrency excess return. The average utility gain for the investor when they use the machine-learning forecast in lieu of the prevailing mean benchmark is then given by

$$\Delta_i = \bar{U}_i^{\text{Compete}} - \bar{U}_i^{\text{Bench}}. \quad (21)$$

After multiplying the average utility gain in Equation (21) by 365, it can be interpreted as the annualized portfolio management fee that the investor would be willing to pay to have access to the information in the competing machine-learning forecast vis-à-vis the prevailing mean benchmark. In this way, Equation (21) measures the economic value of return predictability to the investor.

Table 4 reports annualized average utility gains (in percent) for each of the cryptocurren-

²²We use a value of 0.94 for the decay parameter.

²³We assume that the investor always uses the exponentially weighted moving-average estimator to forecast the variance so that we can isolate the effects of the cryptocurrency excess return forecasts on the investor's utility.

Table 4. Average utility gains

The table reports annualized average utility gains in percent for a mean-variance investor with a coefficient of relative risk aversion of three who allocates daily between the cryptocurrency in the first column and risk-free Treasury bills. The utility gain corresponds to the case where the investor allocates their portfolio based on the forecast of the daily cryptocurrency excess return in the column heading instead of the prevailing mean benchmark forecast. The investor always uses an exponentially weighted moving-average estimator (with a decay parameter of 0.94) to forecast the variance. The start of the out-of-sample period for forecast evaluation is given in the second column; the end of the out-of-sample period is 2021-12-31 for all of the cryptocurrencies.

(1) Cryptocurrency	(2) Out-of-sample start	(3) Linear- ENet	(4) Random forest	(5) XGBoost	(6) Neural network	(7) Ensemble- nonlinear	(8) Ensemble- all
linch	2021-01-24	83.85	48.94	38.50	95.43	72.26	83.10
Aave	2020-11-08	78.83	55.62	108.45	82.93	93.12	97.04
Algorand	2019-07-21	103.87	100.76	101.31	114.88	110.50	110.00
Balancer	2020-07-24	32.23	41.34	66.47	53.68	61.35	60.05
Basic Attention Token	2017-11-04	19.51	32.67	27.24	35.36	46.34	45.88
Bitcoin	2014-01-01	3.86	34.98	111.31	46.16	81.25	61.69
Bitcoin Cash	2017-08-30	10.17	-45.46	-25.09	6.17	-6.00	-0.09
Bitcoin SV	2018-12-14	32.52	12.57	47.81	66.25	55.40	52.83
Cardano	2017-12-30	22.85	33.94	94.08	44.05	71.36	62.51
Chainlink	2017-10-28	17.16	17.71	47.07	17.89	38.14	36.57
Compound	2020-07-17	17.99	16.69	15.04	23.18	19.76	24.44
Crypto.com Coin	2019-04-18	18.12	17.52	50.04	26.92	44.66	39.41
Curve DAO Token	2020-09-13	81.64	25.39	4.20	92.28	48.60	62.93
Dash	2014-03-09	119.27	117.74	149.60	58.13	118.89	135.66
Decentraland	2017-09-23	196.36	185.61	166.24	210.41	202.18	205.58
Decred	2016-06-15	14.24	-0.69	30.99	35.95	32.99	30.84

cies and the different forecasting methods. The second column again gives the start of the out-of-sample period for each cryptocurrency, and the out-of-sample period ends on 2021-12-31 for all of the cryptocurrencies. Of the 246 average utility gains, 228 (93%) are positive.

Table 4 (continued)

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cryptocurrency	Out-of-sample start	Linear- ENet	Random forest	XGBoost	Neural network	Ensemble- nonlinear	Ensemble- all
Dogecoin	2014-02-21	-19.27	-34.90	-18.80	1.15	4.70	6.12
Ethereum	2015-09-06	30.62	35.05	78.47	40.83	77.38	70.87
Ethereum Classic	2016-08-23	90.47	98.83	93.12	99.78	117.06	114.84
FTX Token	2019-09-18	32.54	28.38	23.75	36.71	36.83	36.82
Gnosis	2017-05-31	-17.92	-6.68	26.14	23.85	24.57	14.92
HedgeTrade	2019-12-01	72.55	84.64	-33.34	84.23	77.85	82.45
Huobi Token	2019-04-04	37.82	14.34	34.79	54.72	37.11	40.93
Internet Computer	2021-06-09	109.12	88.22	132.00	77.68	108.81	110.78
Litecoin	2014-01-01	44.75	41.50	79.59	57.62	74.68	70.80
Livepeer	2019-01-18	39.85	-6.86	-34.82	60.13	13.18	31.00
Maker	2018-01-24	-10.27	-5.56	33.99	7.32	25.13	19.02
Neo	2017-08-13	66.28	69.38	90.95	86.65	93.09	89.70
OMGNetwork	2017-08-13	136.43	132.95	183.50	142.58	164.10	164.07
Perpetual Protocol	2021-03-05	103.76	52.38	32.18	115.38	85.14	99.69
Polkadot	2020-09-18	84.13	38.04	38.82	67.25	52.91	70.23
Quant	2019-04-14	23.44	24.51	18.67	27.31	29.36	31.62
Stellar	2015-10-29	44.66	33.47	47.41	27.76	52.18	53.39
SushiSwap	2020-09-30	110.36	100.57	101.35	130.96	112.36	117.77
Synthetix	2020-05-08	44.09	26.19	66.33	42.10	51.81	51.34
Tezos	2018-07-29	12.68	2.23	-8.45	-4.68	7.86	6.78
UMA	2020-10-07	142.24	150.07	101.15	143.89	138.21	143.91
Uniswap	2020-10-17	128.54	98.48	67.42	156.65	104.25	108.82
XRP	2014-09-13	44.05	49.83	103.22	74.23	98.66	91.64
yearn.finance	2020-08-23	150.12	105.39	132.36	178.47	152.23	162.58
Zcash	2016-11-27	13.94	26.60	-3.35	57.43	43.95	43.12

Thus, in the vast majority of cases, the investor benefits from relying on a machine-learning forecast in lieu of the prevailing mean benchmark to guide asset allocation. Furthermore, the

positive gains are extremely large in general. They are typically above 20%, and a number are well above 100%.²⁴ Focusing on the two largest and best-known cryptocurrencies, Bitcoin and Ethereum, the average utility gains range from 3.86% (linear-ENet) to 111.31% (XGBoost) for the former and 30.62% (linear-ENet) to 78.47% (XGBoost) for the latter. Thus, return predictability has substantive economic value for the most familiar cryptocurrencies.

Figure 4 plots log cumulative excess returns for the portfolios based on the ensemble-nonlinear and prevailing mean benchmark forecasts.²⁵ For the vast majority of cryptocurrencies, the cumulative excess return profile for the portfolio based on the ensemble-nonlinear forecast appears clearly superior to that based on the prevailing mean benchmark forecast, consistent with the results in Table 4. Figure 4 provides further evidence of the value of machine-learning forecasts for cryptocurrency return prediction.

5.2. Long-Short Portfolios

Next, we construct a zero-investment long-short portfolio that invests in multiple cryptocurrencies. We form the portfolio by sorting the available cryptocurrencies in our sample using their excess return forecasts based on one of the machine-learning methods. Specifically, at the end of day t , we compute excess return forecasts for all of the available cryptocurrencies for day $t + 1$ using a machine-learning method and sort the cryptocurrencies based on the return forecasts. The portfolio goes long (short) the 30% of cryptocurrencies with the highest (lowest) return forecasts.²⁶ When forming the long and short legs, we consider both value and equal weighting. The out-of-sample period for the long-short portfolio spans January

²⁴Table A1 in the Online Appendix reports average utility gains precluding short sales and assuming a unit transaction cost of 50 basis points for cryptocurrencies. Based on trading cost information for cryptocurrencies reported in Liu et al. (2022), a unit transaction cost of 50 basis points appears relatively large, so the results are likely to be conservative. After precluding short selling and allowing for transaction costs, the average utility gains continue to be positive in the vast majority of cases and generally very large.

²⁵To conserve space, we focus on the ensemble-nonlinear forecast in Figure 4. As shown in Figures A11 to A15 in the Online Appendix, the plots are similar for the other machine-learning forecasts.

²⁶For some days, a tree-based forecast gives the same excess return forecast for all of the available cryptocurrencies, so we cannot sort the cryptocurrencies by the forecasts. For such days, we assume that the portfolio does not invest in cryptocurrencies, so the excess return for the long-short portfolio is zero.

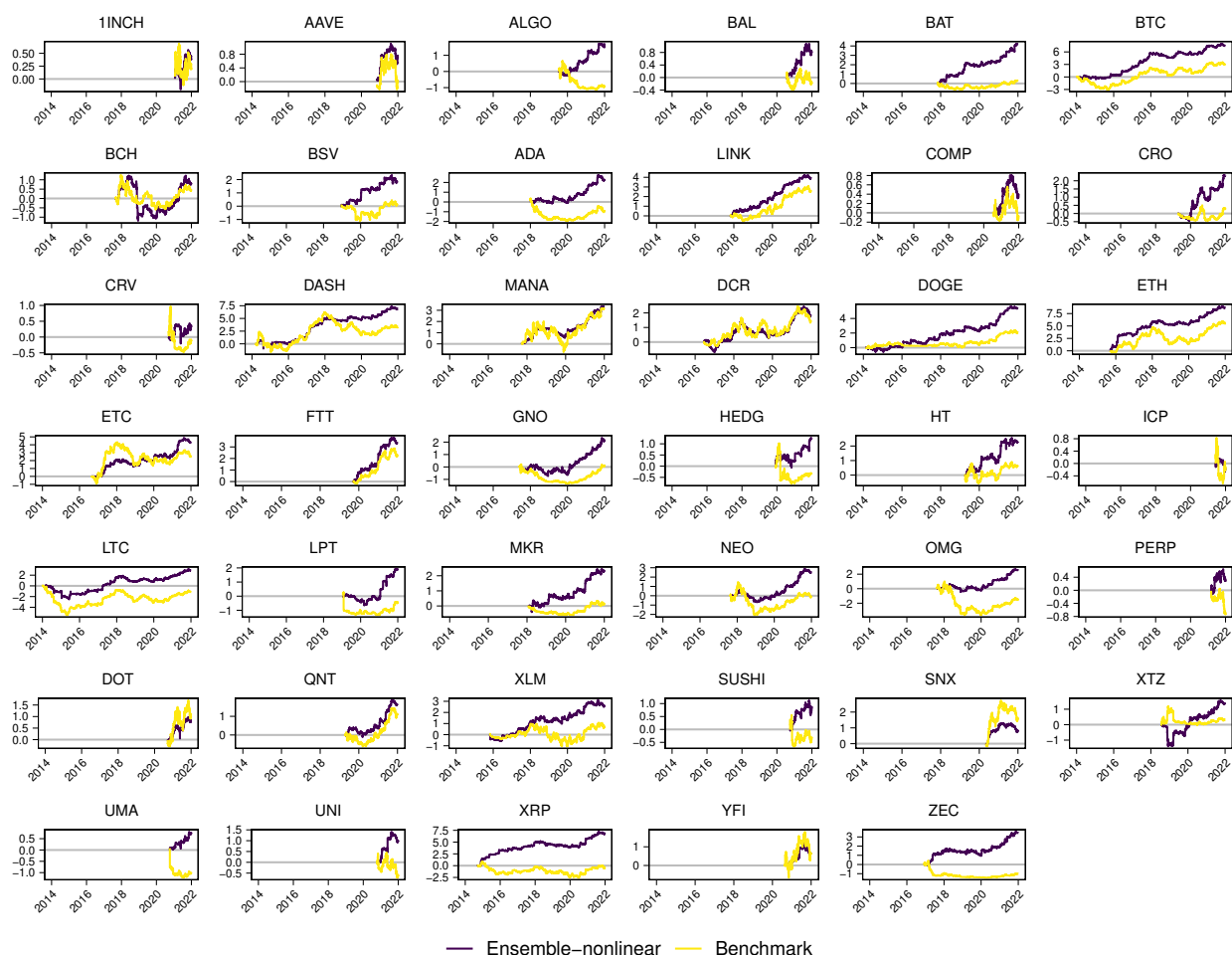


Figure 4. Portfolio log cumulative excess returns

Each panel depicts the portfolio log cumulative excess return for a mean-variance investor with a coefficient of relative risk aversion of three who allocates daily between the cryptocurrency (denoted by its symbol) in the panel heading and risk-free Treasury bills. The investor allocates their portfolio based on the ensemble-nonlinear or prevailing mean benchmark forecast.

1, 2014 to December 31, 2021. As previously mentioned, the beginning of the out-of-sample period coincides with the starting date for [Liu et al. \(2022\)](#) when they construct long-short portfolios based on characteristic sorts to obtain cryptocurrency factors.

Table 5 reports performance metrics for long-short portfolios constructed using the sorted machine-learning cryptocurrency excess return forecasts. The table also reports metrics for

Table 5. Performance metrics

The table reports performance metrics for zero-investment long-short portfolios that invest in multiple cryptocurrencies. The long-short portfolio is constructed by sorting cryptocurrencies according to their excess return forecasts for the available cryptocurrencies on a given day based on the machine-learning method in the first column; the portfolio goes long (short) the 30% of cryptocurrencies with the highest (lowest) excess return forecasts. The long and short legs use value (equal) weighting in Panel A (B). Panel C reports performance metrics for the excess return for a cryptocurrency value-weighted market portfolio that invests in all of the available cryptocurrencies on a given day. Panel D reports performance metrics for the excess return for the CRSP value-weighted aggregate equity market portfolio. The portfolios are scaled to have the same annualized volatility as the equity market portfolio (17.58%). The forecast evaluation period is 2014-01-01 to 2021-12-31.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Forecast	Ann. mean (%)	Ann. Sharpe ratio	Maximum drawdown (%)	Ann. Calmar ratio	Ann. downside risk (%)	Ann. Sortino ratio
Panel A: Value weighting						
Linear-ENet	21.34	1.21	17.81	1.20	11.43	1.87
Random forest	22.54	1.28	18.58	1.21	17.23	1.31
XGBoost	39.00	2.22	12.39	3.15	12.83	3.04
Neural network	18.64	1.06	32.26	0.58	15.84	1.18
Ensemble-nonlinear	37.00	2.10	23.97	1.54	12.66	2.92
Ensemble-all	39.84	2.27	13.40	2.97	10.62	3.75
Panel B: Equal weighting						
Linear-ENet	18.05	1.03	27.83	0.65	13.67	1.32
Random forest	25.23	1.44	14.34	1.76	16.90	1.49
XGBoost	47.20	2.68	9.57	4.93	11.30	4.18
Neural network	20.00	1.14	35.59	0.56	15.84	1.26
Ensemble-nonlinear	45.17	2.57	15.91	2.84	11.58	3.90
Ensemble-all	43.33	2.46	11.95	3.63	10.74	4.04
Panel C: Cryptocurrency market portfolio						
Excess return	19.08	1.09	33.37	0.57	13.67	1.40
Panel D: Equity market portfolio						
Excess return	14.41	0.82	34.35	0.44	15.28	0.94

a cryptocurrency market portfolio based on the available cryptocurrencies in our dataset. Specifically, we compute the excess return for a value-weighted portfolio that invests in

all of the available cryptocurrencies on a given day. In addition, we report metrics for the CRSP value-weighted equity market portfolio based on daily excess return data from Kenneth French's [Data Library](#) for January 1, 2014 to December 31, 2021 (matching the forecast evaluation period). To facilitate comparisons across portfolios, we scale the long-short portfolios based on the machine-learning forecasts and the cryptocurrency market portfolio to have an annualized volatility of 17.58%, matching that for the equity market portfolio.²⁷

The annualized average excess return for the cryptocurrency market portfolio in Panel C is 19.08%, which is more than 450 basis points higher than that for the equity market portfolio in Panel D. With the exception of the neural network (linear-ENet) forecast in Panel A (B), the machine-learning forecasts generate higher average excess returns than the cryptocurrency market portfolio. Indeed, the average excess returns are approximately twice as large or larger for the XGBoost, ensemble-nonlinear, and ensemble-all forecasts for both value and equal weighting. The annualized Sharpe ratio is 1.09 (0.82) for the cryptocurrency (equity) market portfolio. Nearly all of the long-short portfolios based on the machine-learning forecasts generate larger Sharpe ratios than the cryptocurrency market portfolio, and numerous forecasts produce Sharpe ratios that are well above two.

The maximum drawdown for the cryptocurrency market portfolio is 33.37%, which is just below that (34.35%) for the equity market portfolio. With one exception (the neural network forecast in Panel B), the maximum drawdowns for the long-short portfolios are below that for the cryptocurrency market portfolio; in a number of cases, they are less than half as large. The average excess returns and maximum drawdowns typically translate into large Calmar ratios in the fifth column of Table 5. The annualized Calmar ratio for the cryptocurrency (equity) market portfolio is 0.57 (0.44), while the Calmar ratios are much higher for the XGBoost, ensemble-nonlinear, and ensemble-all forecasts in Panel A (B), with

²⁷Equity market excess return observations are not available for weekends and holidays. The performance metrics for the equity market portfolio in Table 5 are computed using the available observations.

values of 3.15, 1.54, and 2.97 (4.93, 2.84, and 3.63), respectively.

In many cases, the annualized downside risks for the long-short portfolios in Panels A and B are smaller than those for the cryptocurrency and equity market portfolios (13.67% and 15.28%, respectively). The average excess returns and downside risks lead to annualized Sortino ratios in the last column of Table 5 that are often considerably larger in Panels A and B than those for the cryptocurrency and equity market portfolios (1.40 and 0.94, respectively), especially for the XGBoost, ensemble-nonlinear, and ensemble-all forecasts. In general, Table 5 indicates that the machine-learning forecasts generate substantive improvements in portfolio performance relative to passively holding a value-weighted cryptocurrency or equity market portfolio.²⁸

We further examine whether the long-short portfolios based on sorted machine-learning forecasts of cryptocurrency returns earn risk-adjusted returns. Specifically, we test whether the portfolios generate significant alpha in the context of the Liu et al. (2022) cryptocurrency three-factor model. From a list of leading equity market characteristics, Liu et al. (2022) compute cryptocurrency analogs and find that a three-factor model comprised of market, size, and momentum factors performs well in accounting for cross-sectional cryptocurrency excess returns. We construct cryptocurrency market, size, and momentum factors for the available cryptocurrencies in our dataset along the lines of Liu et al. (2022). The market factor (CMKT) is the value-weighted excess return for the cryptocurrency market portfolio previously described. The size factor (CSMB) is formed by sorting the available cryptocurrencies according to their market capitalization at the end of the previous day; the size factor is the return on a portfolio that goes long (short) the 30% of cryptocurrencies with the smallest (largest) market capitalization, where the long and short legs use value weighting. For the momentum factor (CMOM), we sort cryptocurrencies according to their cumulative returns

²⁸Table A2 in the Online Appendix reports performance metrics for the long legs of the long-short portfolios (thereby precluding short selling) and assuming a unit transaction cost of 50 basis points for cryptocurrencies. The portfolios' long legs with transaction costs generally continue to exhibit impressive performance.

over the previous 21 days and compute the return on a portfolio that goes long (short) the 30% of cryptocurrencies with the highest (lowest) cumulative returns, where we again use value weighting in the long and short legs.

Multifactor model estimation results are presented in Table 6.²⁹ For value weighting in Panel A, all of the long-short portfolios based on the machine-learning forecasts evince statistically significant exposures to the market factor, but the estimated betas are fairly moderate, ranging from 0.06 (random forest) to 0.24 (XGBoost). All of the long-short portfolios also have significantly negative exposures to the size factor, so the portfolios have a large-cap tilt. The exposures to the momentum factor are all significantly positive, so the portfolios have a momentum tilt as well. The significant factor exposures, however, are typically unable to explain the average excess returns for the long-short portfolios, as the portfolios deliver significant risk-adjusted excess returns. In particular, the portfolio alphas are significant at the 5% level for the neural network forecast and the 1% level for the XGBoost, ensemble-nonlinear, and ensemble-all forecasts; for the latter three forecasts, the annualized alphas are 208%, 196%, and 186%, respectively.

The results for long-short portfolios based on equal weighting in Panel B are similar to those in Panel A, in that the portfolios exhibit significantly positive (negative) exposures to the market and momentum (size) factors and typically generate significant alphas. With the exception of the linear-ENet forecast, the alphas in Panel B are larger than the corresponding values in Panel A. Again with the exception of the linear-ENet forecast, all of the forecasts produce significant alphas (at the 5% or 1% level) in Panel B, with the XGBoost, ensemble-nonlinear, and ensemble-all forecasts delivering alphas of 248%, 240%, and 204%, respectively.

Complementing the evidence in Section 4.2, Table 6 suggests that nonlinearities are important for forecasting cryptocurrency excess returns. Long-short portfolios based on the linear-ENet forecast fail to produce significant alpha. In contrast, with the exception of

²⁹The long-short portfolio volatilities are not scaled in Table 6.

Table 6. Multifactor model estimation results

The table reports multifactor model estimation results for zero-investment long-short portfolios that invest in multiple cryptocurrencies. The long-short portfolio is constructed by sorting cryptocurrencies according to their excess return forecasts for the available cryptocurrencies in our dataset on a given day based on the machine-learning method in the column heading; the portfolio goes long (short) the 30% of cryptocurrencies with the highest (lowest) excess return forecasts. The long and short legs use value (equal) weighting in Panel A (B). The multifactor model is comprised of cryptocurrency market (CMKT), size (CSMB), and momentum (CMOM) factors. The forecast evaluation period is 2014-01-01 to 2021-12-31. Standard errors are in parentheses; *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Coefficient	Linear- ENet	Random forest	XGBoost	Neural network	Ensemble- nonlinear	Ensemble- all
Panel A: Value weighting						
$\hat{\alpha}$ (annualized %)	37.65 (43.47)	46.97 (29.07)	207.60*** (48.25)	91.50** (44.31)	196.00*** (48.79)	185.63*** (44.07)
$\hat{\beta}_{\text{CMKT}}$	0.16*** (0.03)	0.06*** (0.02)	0.24*** (0.03)	0.15*** (0.03)	0.22*** (0.03)	0.23*** (0.03)
$\hat{\beta}_{\text{CSMB}}$	-0.12*** (0.02)	-0.15*** (0.02)	-0.48*** (0.03)	-0.15*** (0.02)	-0.44*** (0.03)	-0.43*** (0.02)
$\hat{\beta}_{\text{CMOM}}$	0.65*** (0.02)	0.27*** (0.01)	0.31*** (0.02)	0.09*** (0.02)	0.32*** (0.02)	0.39*** (0.02)
R^2 (%)	32.83	16.50	16.33	2.58	15.09	21.24
Panel B: Equal weighting						
$\hat{\alpha}$ (annualized %)	29.46 (43.35)	55.17** (26.84)	247.55*** (42.57)	99.57** (41.83)	240.12*** (43.35)	203.98*** (41.16)
$\hat{\beta}_{\text{CMKT}}$	0.06** (0.03)	0.04** (0.02)	0.21*** (0.03)	0.09*** (0.03)	0.17*** (0.03)	0.18*** (0.03)
$\hat{\beta}_{\text{CSMB}}$	-0.16*** (0.02)	-0.20*** (0.02)	-0.59*** (0.02)	-0.20*** (0.02)	-0.56*** (0.02)	-0.54*** (0.02)
$\hat{\beta}_{\text{CMOM}}$	0.51*** (0.02)	0.23*** (0.01)	0.23*** (0.02)	0.06*** (0.02)	0.23*** (0.02)	0.30*** (0.02)
R^2 (%)	22.80	16.68	20.92	2.88	18.59	22.16

the random forest in Panel A, the other forecasts—all of which incorporate nonlinearities in the prediction models—generate significant alpha. Indeed, the alphas are often much larger

in magnitude for forecasts that incorporate nonlinearities. We investigate nonlinearities in marginal predictive relations in fitted models underlying the machine-learning forecasts in Section 6.2.

6. Model Interpretation

Sections 4 and 5 show that machine-learning methods based on a rich set of predictors produce significant excess return predictability—both statistically and economically—for a large number of cryptocurrencies. In this section, we analyze the role of the predictors in fitted models that produce the forecasts, which allows us to glean insight into the economic sources of cryptocurrency return predictability. We use model-interpretation tools based on Shapley (1953) values. Utilizing the analogy between players in a cooperative game earning a payoff and predictors in a model generating a forecast, Štrumbelj and Kononenko (2010, 2014) and Lundberg and Lee (2017) develop Shapley-based measures for interpreting fitted machine-learning models. Because Shapley values have a set of appealing properties, they are generally viewed as providing the best basis for interpreting fitted models (Molnar, 2023). We use Shapley-based measures to assess variable (or predictor) importance as well as the strength of nonlinearities in marginal predictive relations in fitted models. We compute Shapley-based measures for the fitted models that generate the final set of daily cryptocurrency excess return forecasts (for December 31, 2021), as these are the fitted models based on the largest training sample of panel data in our application.

6.1. Variable Importance

First, we compute Shapley-based variable-importance metrics for the fitted models for each of the predictors. Figure 5 reports results for the individual machine-learning models, while Figure 6 depicts results for the ensembles. To facilitate interpretation of the results, we combine variable-importance measures across the two transformations (log or simple deviations from 7- and 28-day moving averages), as indicated by “(7,28)” appended to the predictor

abbreviations in the figure. We also combine the variable-importance measures for the two time-series momentum predictors in levels (TSM7 and TSM28). For a given model, we scale the variable-importance measures to sum to 100.

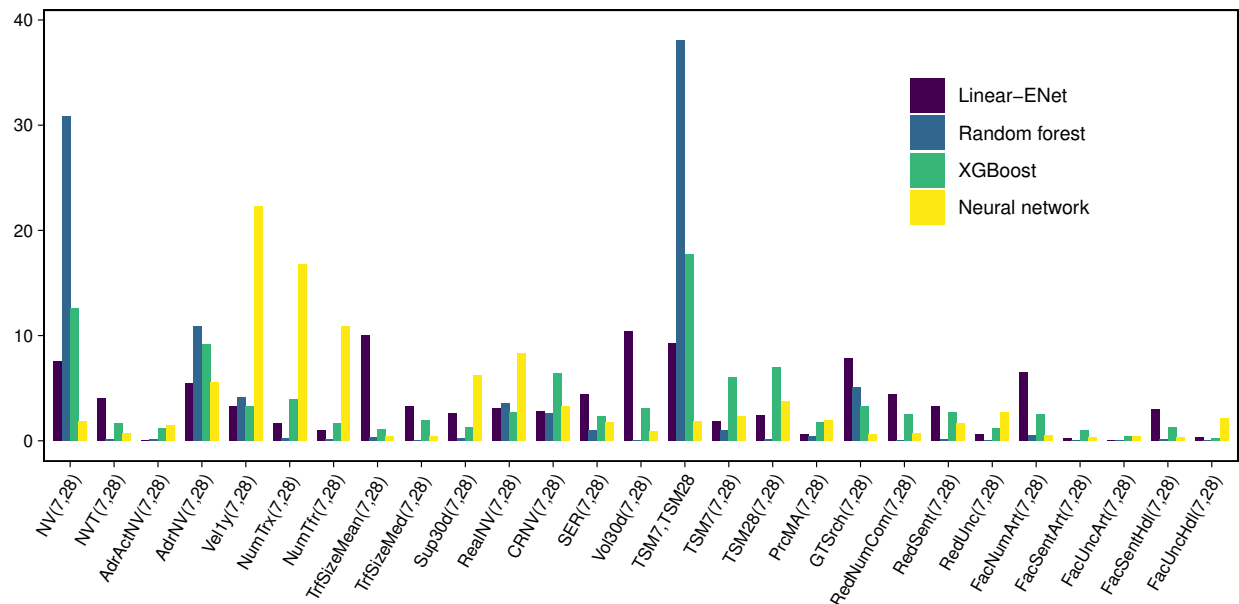


Figure 5. Variable importance

The figure depicts Shapley-based variable-importance (VI) measures for the predictors in the final fitted linear-ENet, random forest, XGBoost, and neural network prediction models for cryptocurrency excess returns. The predictors are denoted by their abbreviations in Table 2. We combine the VI measures for TSM7 and TSM28 in levels; we combine the VI measures for each of the predictors when they are transformed using log or simple deviations from 7- and 28-day moving averages, as indicated by “(7,28).” The VI measures are scaled to sum to 100.

Time-series momentum (TSM7, TSM28) from the Market group in Table 2 stands out as an important predictor for the fitted linear-ENet, random forest, and XGBoost models—especially the random forest—in Figure 5. This finding aligns with Liu and Tsyvinski (2021), who find significant evidence of time-series momentum in cryptocurrency returns on an in-sample basis, as well as Moskowitz et al. (2012), who detect time-series momentum in equity index, currency, commodity, and bond futures returns. Our results provide out-of-sample evidence of the relevance of time-series momentum for cryptocurrency return predictability.

Our results also appear consistent with the inclusion of the momentum factor (CMOM) in the [Liu et al. \(2022\)](#) cryptocurrency three-factor model as well as the momentum effect in equities ([Jegadeesh and Titman, 1993](#)).

From the Network group in Table 2, network value (NV; i.e., market capitalization) is also quite important for the linear-ENet, random forest, and XGBoost models in Figure 5. The importance of network value is consistent with the size factor (CSMB) included in the cryptocurrency three-factor model of [Liu et al. \(2022\)](#). The address-to-network value ratio (AdrNV), also from the Network group, is relatively important for all of the individual fitted models. The importance of the address-to-network value ratio complements [Liu et al. \(2023\)](#), who find evidence of a value effect in cross-sectional cryptocurrency returns. The leading roles played by network value and the address-to-network value ratio in the fitted models for predicting cryptocurrency returns are reminiscent of the well-known size and value effects (e.g., [Basu, 1977](#); [Banz, 1981](#)), respectively, in the equity market.

With respect to the Online Activity group in Table 2, Google Trends search volume (GTSrch) exhibits fairly substantive importance for the linear-ENet, random forest, and XGBoost models, so online activity appears relevant for anticipating cryptocurrency returns in terms of internet searches. The number of Reddit comments (RedNumCom) and the number of Factiva articles (FacNumArt) are also reasonably important for the linear-ENet and XGBoost models, along with the Reddit and Factiva sentiment scores (RedSent and FacSentHdl, respectively).

Relative to the other models, the fitted neural network places considerably more importance on a number of predictors from the Transactions group in Table 2 relating to network activity. Such predictors include velocity (Velly), the number of transactions (NumTrx), the number of transfers (NumTfr), active supply (Sup30d), and realized network value (RealNV). Also from the Transactions group, XGBoost identifies the current-to-realized ratio (CRNV) as relevant, while the linear-ENet points to volatility (Vol30d) from the Market group as important.

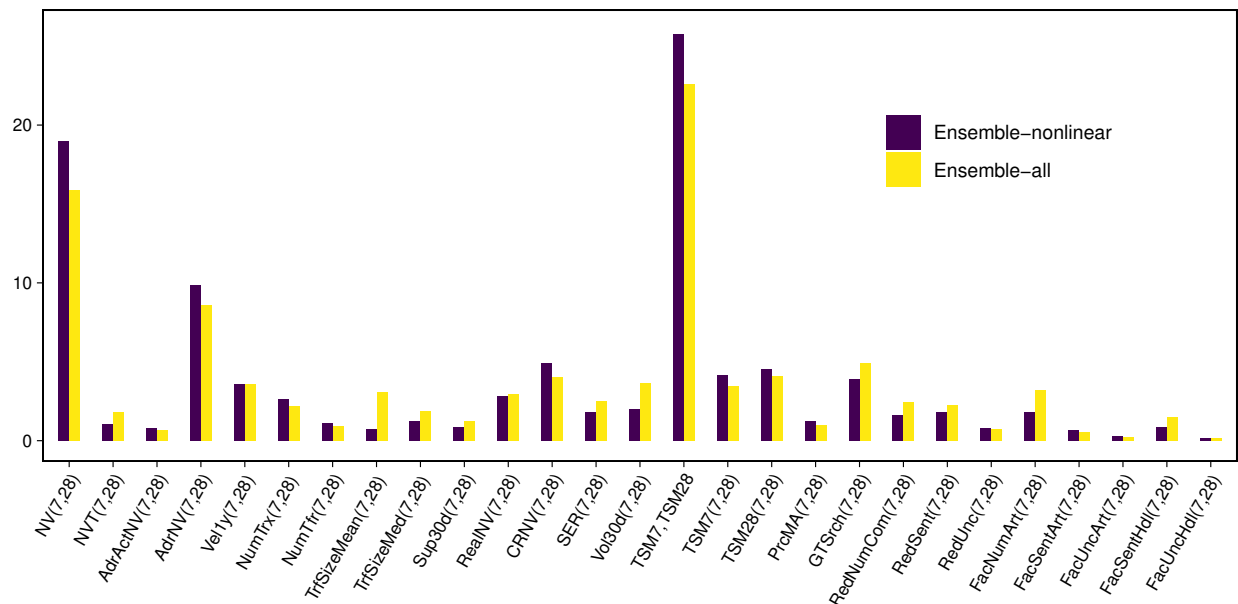


Figure 6. Variable importance: Ensembles

The figure depicts Shapley-based variable-importance (VI) measures for the predictors in the final fitted prediction models for the ensemble-nonlinear and ensemble-all forecasts for cryptocurrency excess returns. The predictors are denoted by their abbreviations in Table 2. We combine the VI measures for TSM7 and TSM28 in levels; we combine the VI measures for each of the predictors when they are transformed using log or simple deviations from 7- and 28-day moving averages, as indicated by “(7,28).” The VI measures are scaled to sum to 100.

Turning to the ensemble forecasts, Figure 6 continues to highlight time-series momentum, network value, and the address-to-network value ratio as important predictors. Indeed, they clearly stand out as the most relevant predictors for cryptocurrency returns with regard to the ensemble-nonlinear and ensemble-all forecasts. From the Transactions group, velocity, the number of transactions, realized network value, and the current-to-realized ratio appear relatively important for both ensembles. From the Online Activity group, Google Trends search volume also evinces considerable importance for both ensembles in Figure 6.

Overall, variable-importance analysis pinpoints time-series momentum, network value, and the address-to-network value ratio as key drivers of future cryptocurrency excess returns. Network value and the address-to-network value ratio can be viewed as cryptocurrency fun-

damentals, indicating the relevance of fundamentals for cryptocurrency return prediction. The importance of a number of measures of network activity further supports the relevance of cryptocurrency fundamentals. At the same time, the leading role played by time-series momentum indicates that trends potentially unrelated to fundamentals exert a substantial effect on cryptocurrency values. The relevance of Google Trends search volume also identifies a substantive role for investor attention via online activity.

Economically, it appears that both rational time-varying risk premia and mispricing play important roles in cryptocurrency return predictability. Because the cryptocurrency market is relatively new, mispricing due to investors still learning about the market would not be very surprising. It will be interesting to see whether return predictability related to mispricing diminishes if cryptocurrencies become more important as an asset class and institutional investors participate more actively in the cryptocurrency market. Limits to arbitrage (Shleifer and Vishny, 1997) provide another source of mispricing. If limits to arbitrage are pervasive and difficult to overcome in the cryptocurrency market, then return predictability via mispricing can persist, even as investors learn more about the market. Of course, we cannot know the true underlying cause or causes of return predictability. Nevertheless, our results suggest that cryptocurrency return predictability reflects both rational time-varying risk premia and mispricing.

It is interesting to note that while network value and time-series momentum are identified as important predictors in Figures 5 and 6 and the long-short portfolios in Table 6 exhibit significant exposures to the cryptocurrency size and momentum factors, many of the long-short portfolios still generate statistically and economically significant alpha.³⁰ This indicates that the machine-learning forecasts process the information in network value and time-series momentum in a manner that adds substantial value relative to the information contained in the cryptocurrency size and momentum factor portfolios.

³⁰Indeed, the exposures to the cryptocurrency size factor are significantly negative in Table 6, which (all else equal) has the effect of increasing alpha.

6.2. Shapley-Value Plots

Next, we depict Shapley-value plots for the predictors to provide perspective on the importance of nonlinearities in the fitted forecasting models. For each observation in the training sample, the Shapley value measures the contribution of a given predictor to the predicted target value for that observation.³¹ Shapley-value plots give a sense of how a fitted model's prediction changes as a predictor changes. As such, they are informative for helping to identify nonlinearities in fitted prediction functions.

To conserve space, Figure 7 depicts Shapley-value plots for the ensembles. Figure A16 in the Online Appendix provides plots for the linear-ENet, random forest, XGBoost, and neural network models.³² Figure 7 reveals important nonlinearities for many predictors for both ensembles. Consider, for example, 7-day time-series momentum (TSM7). For its lowest values, which are negative, the prediction is often positive and large in magnitude, consistent with a type of reversal effect in this region. The prediction is then relatively stable until we reach the highest TSM7 values, where the slope becomes sizably negative.

With respect to network value in log deviation from a 7-day moving average (NV(7)), both ensembles evince strong threshold effects. Specifically, the prediction is approximately 4% until NV(7) reaches around -2.5% , where the prediction falls to near zero and remains there. Threshold effects are also evident for the ensembles for the address-to-network value ratio in log deviation from a 7-day moving average (AdrNV(7)), as the prediction increases sharply when AdrNV(7) is near 2%. The variable-importance measures in Figure 5 point to the relevance of momentum, size, and value effects for cryptocurrency return predictability, while the Shapley-value plots in Figure 7 indicate that these effects are often characterized by important nonlinearities in predictive relations.

From the Online Activity group, Google Trends search volume in log deviation from a

³¹Because there are a very large number of observations (55,655) in the final training sample of panel data that is used to fit the models, we smooth the Shapley-value plots using 200 bins.

³²By construction, the Shapley-value plots are linear for the linear-ENet.

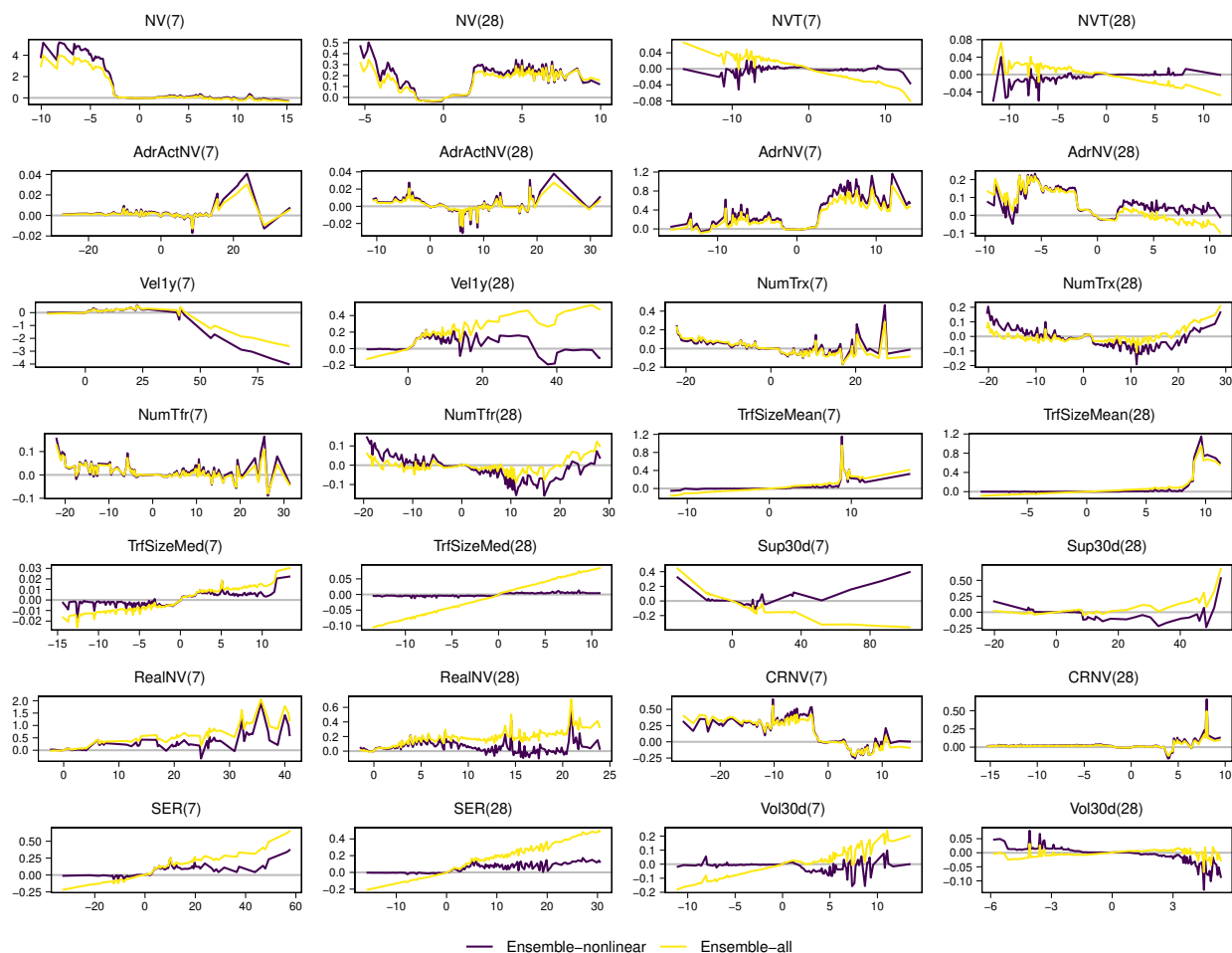


Figure 7: Shapley values: Ensembles

Each panel depicts Shapley-value plots for the predictor (denoted by its abbreviation in Table 2) in the panel heading in the final fitted prediction models for the ensemble-nonlinear and ensemble-all forecasts for cryptocurrency excess returns. Parentheses indicate that the predictor is transformed using the log or simple deviation from a 7- or 28-day moving average. The Shapley-value plots are smoothed using 200 bins.

7-day moving average (GTSrch(7)) displays a substantial threshold effect near 4% corresponding to an increase in the predicted excess return. From the Transaction group, the Shapley-value plots for the number of transactions and number of transfers in log deviations from 28-day moving averages (NumTrx(28) and NumTrf(28), respectively) exhibit distinct

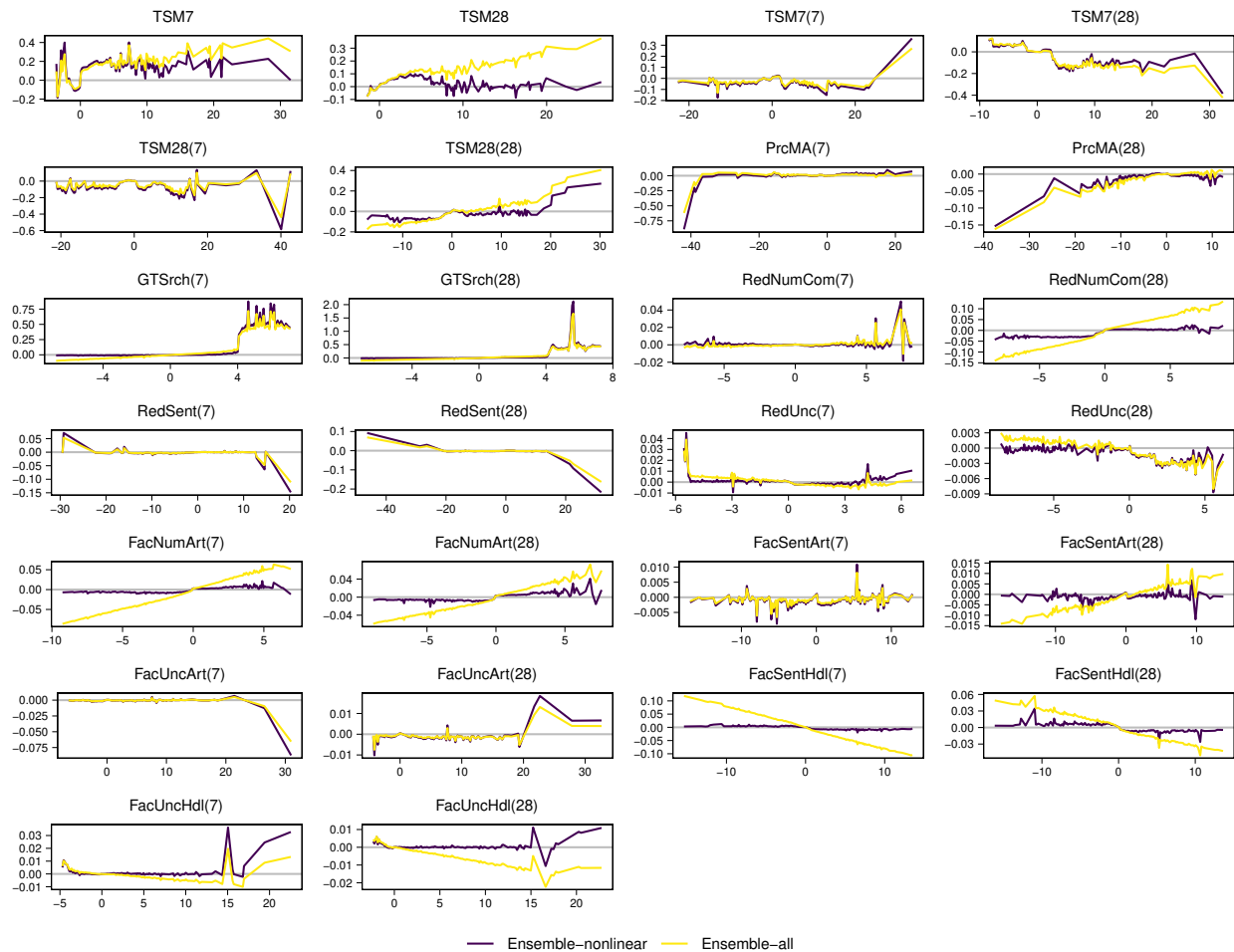


Figure 7 (continued)

U-shaped patterns.

7. Conclusion

We provide an in-depth investigation of the out-of-sample predictability of a large number of daily cryptocurrency excess returns. Our analysis uses a large set of predictors from diverse categories and employs modern machine-learning methods to generate return forecasts. Incorporating the information in the predictors via machine learning substantially improves out-of-sample forecasts of daily cryptocurrency excess returns in terms of out-of-sample MSE.

In addition, relying on machine-learning forecasts to guide asset allocation provides substantive economic value: machine-learning forecasts generate large utility gains for an investor who allocates between individual cryptocurrencies and Treasury bills, and long-short portfolios that go long (short) cryptocurrencies with the highest (lowest) return forecasts typically produce impressive Sharpe, Calmar, and Sortino ratios and deliver sizable alphas in the context of the [Liu et al. \(2022\)](#) cryptocurrency three-factor model. Machine-learning models that allow for nonlinearities are especially useful for producing out-of-sample gains, as they generate significant improvements in performance vis-à-vis linear models. A leading feature of cryptocurrencies is their large swings in value, and we find that our machine-learning forecasts are proficient at predicting cryptocurrency returns around such swings.

We employ Shapley values to explore the economic underpinnings of cryptocurrency return predictability. According to Shapley-based variable-importance analysis, time-series momentum, network value, and the address-to-network value ratio stand out as key drivers of cryptocurrency return predictability. The importance of these predictors is reminiscent of the well-known momentum, size, and value effects in the equity market. Among other predictors that appear important are those relating to investor attention as measured via online activity, including Google Trends search volume. Overall, we find that predictors closely linked to fundamentals (e.g., network value and the address-to-network value ratio) as well as some less obviously tied to fundamentals (e.g., time-series momentum and Google Trends search volume) play leading roles in cryptocurrency return predictability. In line with our finding that machine-learning methods that accommodate nonlinearities generally outperform linear models, Shapley-value plots reveal substantive nonlinearities in predictive relations for fitted machine-learning models. Nonlinearities are clearly evident in predictive relations for the most important predictors and frequently take the form of strong threshold effects.

At the deepest level, like other asset classes, cryptocurrency return predictability is due to one or more of three factors. First, return predictability can reflect rational time-varying risk

premia. Second, it can emanate from mispricing related to factors such as investor learning and limits to arbitrage in the cryptocurrency market. Third, it can result from neglected trading costs faced by investors. We provide strong evidence that return predictability exists across a broad range of cryptocurrencies, in terms of both statistical accuracy and economic value. Using Shapley-based model-interpretation tools, we find that predictors closely tied to fundamentals as well as those that appear more weakly linked to fundamentals are both important determinants of cryptocurrency return predictability. Although it is not possible to definitively determine the ultimate cause or causes of return predictability, our results suggest that rational time-varying risk premia and mispricing both play significant roles (while neglected trading costs are a less likely factor) in generating cryptocurrency return predictability. Given that the cryptocurrency market is still relatively young, as the market evolves and investors become more familiar with this new asset class, it will be interesting to monitor the future evolution of cryptocurrency return predictability. Along the lines of [McLean and Pontiff \(2016\)](#), this can help to shed further light on the ultimate sources of return predictability. In addition, an interesting avenue for future research is to investigate the sources of the nonlinearities in cryptocurrency return prediction, including the threshold effects associated with a number of important predictors.

References

- Akiba, T., S. Sano, T. Yanase, T. Ohta, and M. Koyama (2019). Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.
- Avramov, D., S. Cheng, and L. Metzker (2023). Machine learning versus economic restrictions: Evidence from stock return predictability. *Management Science* 69(5), 2587–2619.
- Banz, R. W. (1981). The relationship between return and market value of common stocks. *Journal of Financial Economics* 9(1), 3–18.

- Basu, S. (1977). Investment performance of common stocks in relation to their price-earnings ratios: A test of the efficient market hypothesis. *Journal of Finance* 32(3), 663–682.
- Biais, B., C. Bisiere, M. Bouvard, C. Casamatta, and A. J. Menkveld (2023). Equilibrium Bitcoin pricing. *Journal of Finance* 78(2), 967–1014.
- Breiman, L. (1996). Bagging predictors. *Machine Learning* 24(1), 123–140.
- Breiman, L. (1997). Arcing the edge. Technical Report 486, Statistics Department, University of California, Berkeley.
- Breiman, L. (2001). Random forests. *Machine Learning* 45(1), 5–32.
- Breiman, L., J. Friedman, C. J. Stone, and R. A. Olshen (1984). *Classification and Regression Trees*. New York: CRC Press.
- Campbell, J. Y. and S. B. Thompson (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies* 21(4), 1509–1531.
- Cheah, J. E.-T., D. Luo, Z. Zhang, and M.-C. Sung (2020). Predictability of Bitcoin returns. *European Journal of Finance* 28(1), 1–20.
- Chen, T. and C. Guestrin (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.
- Chen, W., H. Xu, L. Jia, and Y. Gao (2021). Machine learning model for Bitcoin exchange rate prediction using economic and technology determinants. *International Journal of Forecasting* 37(1), 28–43.
- Chinco, A., A. D. Clark-Joseph, and M. Ye (2019). Sparse signals in the cross-section of returns. *Journal of Finance* 74(1), 449–492.

- Cong, L. W., Y. Li, and N. Wang (2021). Tokenomics: Dynamic adoption and valuation. *Review of Financial Studies* 34(3), 1105–1155.
- Cybenko, G. (1989). Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals, and Systems* 2(4), 303–314.
- Detzel, A. L., H. Liu, J. Strauss, G. Zhou, and Y. Zhu (2021). Learning and predictability via technical analysis: Evidence from Bitcoin and stocks with hard-to-value fundamentals. *Financial Management* 50(1), 107–137.
- Diebold, F. X. and R. S. Mariano (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics* 13(3), 253–263.
- Dong, X., Y. Li, D. E. Rapach, and G. Zhou (2022). Anomalies and the expected market return. *Journal of Finance* 77(1), 639–681.
- Fama, E. F. and K. R. French (1989). Business conditions and the expected returns on stocks and bonds. *Journal of Financial Economics* 25(1), 23–49.
- Filippou, I., D. E. Rapach, M. P. Taylor, and G. Zhou (2023). Economic fundamentals and short-run exchange rate prediction: A machine learning perspective. Working Paper (available at <https://ssrn.com/abstract=3455713>).
- Flitter, E. and D. Yaffe-Bellany (2023). FTX founder gamed markets, crypto rivals say. *New York Times*, January 18, 2023.
- Freyberger, J., A. Neuhierl, and M. Weber (2020). Dissecting characteristics nonparametrically. *Review of Financial Studies* 33(5), 2326–2377.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics* 29(5), 1189–1232.

- Friedman, J. H. (2002). Stochastic gradient boosting. *Computational Statistics & Data Analysis* 38(4), 367–378.
- Funahashi, K.-I. (1989). On the approximate realization of continuous mappings by neural networks. *Neural Networks* 2(3), 183–192.
- Goodfellow, I., Y. Bengio, and A. Courville (2016). *Deep Learning*. Boston: MIT Press.
- Goyal, A. and I. Welch (2003). Predicting the equity premium with dividend ratios. *Management Science* 49(5), 639–654.
- Goyal, A. and I. Welch (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies* 21(4), 1455–1508.
- Gradojevic, N., D. Kukolj, R. Adcock, and V. Djakovic (2023). Forecasting Bitcoin with technical analysis: A not-so-random forest? *International Journal of Forecasting* 39(1), 1–17.
- Griffin, J. M. and A. Shams (2020). Is Bitcoin really untethered? *Journal of Finance* 75(4), 1913–1964.
- Gu, S., B. Kelly, and D. Xiu (2020). Empirical asset pricing via machine learning. *Review of Financial Studies* 33(5), 2223–2273.
- Harvey, C. R., T. A. Zeid, T. Draaisma, M. Luk, H. Neville, A. Rzym, and O. V. Hemert (2022). An investor’s guide to crypto. *Journal of Portfolio Management* 49(1), 146–171.
- Hoerl, A. E. and R. W. Kennard (1970). Ridge regression: Applications to nonorthogonal problems. *Technometrics* 12(1), 69–82.
- Hornik, K. (1991). Approximation capabilities of multilayer feedforward networks. *Neural Networks* 4(2), 251–257.

- Hornik, K., M. Stinchcombe, and H. White (1989). Multilayer feedforward networks are universal approximators. *Neural Networks* 2(5), 359–366.
- Huang, J.-Z., W. Huang, and J. Ni (2019). Predicting Bitcoin returns using high-dimensional technical indicators. *Journal of Finance and Data Science* 5(3), 140–155.
- Jegadeesh, N. and S. Titman (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance* 48(1), 65–91.
- Kingma, D. P. and J. Ba (2015). Adam: A method for stochastic optimization. In *Third Annual International Conference on Learning Representations*.
- Kozak, S., S. Nagel, and S. Santosh (2020). Shrinking the cross-section. *Journal of Financial Economics* 135(2), 271–292.
- Liu, Y. and A. Tsyvinski (2021). Risks and returns of cryptocurrency. *Review of Financial Studies* 34(6), 2689–2727.
- Liu, Y., A. Tsyvinski, and X. Wu (2022). Common risk factors in cryptocurrency. *Journal of Finance* 77(2), 1133–1177.
- Liu, Y., A. Tsyvinski, and X. Wu (2023). Accounting for cryptocurrency value. Working Paper (available at <https://ssrn.com/abstract=3951514>).
- Loughran, T. and B. McDonald (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance* 66(1), 35–65.
- Lundberg, S. M. and S.-I. Lee (2017). A unified approach to interpreting model predictions. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*.

- Maas, A. L., A. Y. Hannun, and A. Y. Ng (2013). Rectifier nonlinearities improve neural network acoustic models. In *Proceedings of the 30th International Conference on Machine Learning*.
- Makarov, I. and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135(2), 293–319.
- Martin, I. and S. Nagel (2022). Market efficiency in the age of big data. *Journal of Financial Economics* 145(1), 154–177.
- McLean, R. D. and J. Pontiff (2016). Does academic research destroy stock return predictability? *Journal of Finance* 71(1), 5–31.
- Molnar, C. (2023). *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable* (Second ed.). Independently published.
- Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen (2012). Time series momentum. *Journal of Financial Economics* 104(2), 228–250.
- Nagel, S. (2021). *Machine Learning in Asset Pricing*. Princeton, NJ: Princeton University Press.
- Porter, M. F. (1980). An algorithm for suffix stripping. *Program* 14(3), 130–137.
- Prasad, E. S. (2021). *The Future of Money*. Cambridge, MA: Harvard University Press.
- Rapach, D. E., J. K. Strauss, and G. Zhou (2010). Out-of-sample equity premium prediction: Combination forecasts and links to the real economy. *Review of Financial Studies* 23(2), 821–862.
- Rolnick, D. and M. Tegmark (2018). The power of deeper networks for expressing natural functions. In *Proceedings of the Sixth Annual International Conference on Learning Representations*.

- Shapley, L. S. (1953). A value for n -person games. *Contributions to the Theory of Games* 2(28), 307–317.
- Shleifer, A. and R. W. Vishny (1997). The limits of arbitrage. *Journal of Finance* 52(1), 35–55.
- Srivastava, N., G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research* 15(56), 1929–1958.
- Štrumbelj, E. and I. Kononenko (2010). An efficient explanation of individual classifications using game theory. *Journal of Machine Learning Research* 11(1), 1–18.
- Štrumbelj, E. and I. Kononenko (2014). Explaining prediction models and individual predictions with feature contributions. *Knowledge and Information Systems* 41(1), 647–665.
- Tibshirani, R. (1996). Regression shrinkage and selection via the LASSO. *Journal of the Royal Statistical Society. Series B (Methodological)* 58(1), 267–288.
- Timmermann, A. (2006). Forecast combinations. In G. Elliott, C. W. J. Granger, and A. Timmermann (Eds.), *Handbook of Economic Forecasting*, Volume 1, pp. 135–196. Amsterdam: Elsevier.
- West, K. D. (1996). Asymptotic inference about predictive ability. *Econometrica* 64(5), 1067–1084.
- Zou, H. and T. Hastie (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society. Series B (Statistical Methodology)* 67(2), 301–320.